App Building with .NET MAUI in .NET 9

David Ortinau

@davidortinau



WHO IS

David Ortinau

Principal Product Manager

St. Louis, Missouri USA

Help customers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform – aka .NET MAUI.

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age. Drives a Rivian.



Who are you?

What questions do you want answered?

What are you hoping to see?

Microsoft's Investments in Client Development

Windows Native





WinUI3

.NET

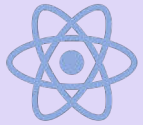
WinForms

.NET

WPF

Cross Platform Native





React Native



.NET MAUI

Hybrid





Blazor Hybrid



WebView2

Web



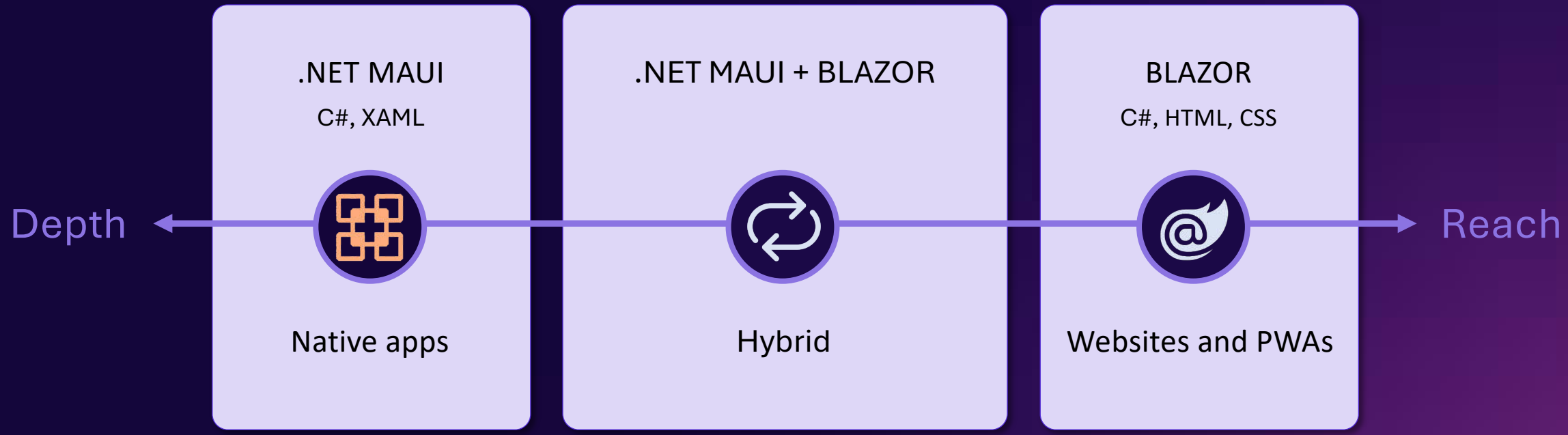
PWA

Progressive Web Apps



Blazor Web

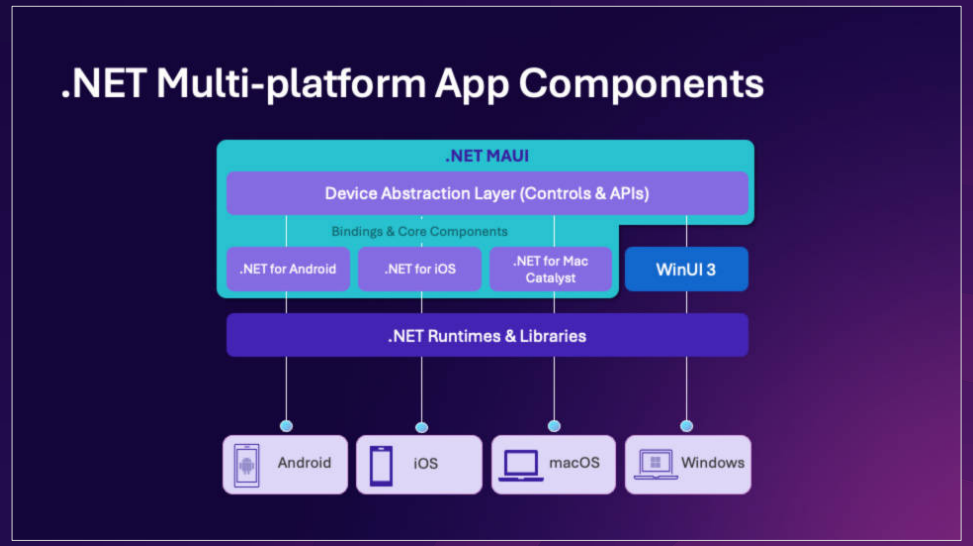
.NET offers choice for client app developers



.NET MAUI

Build native, multi-platform apps all in one framework.

- Cross-platform native mobile and desktop apps for Android, iOS, Mac, Windows
- Modern native client stack for .NET
- Single project, single codebase
- Native controls and styling
- Access over 60 common device APIs
- Access platform specific APIs if needed
- Built on top of .NET, with access to Base Class Libraries
- App store distribution and reach
- Productivity tools for Visual Studio and VS Code
- Platform is open source on GitHub
- Healthy ecosystem of controls and components

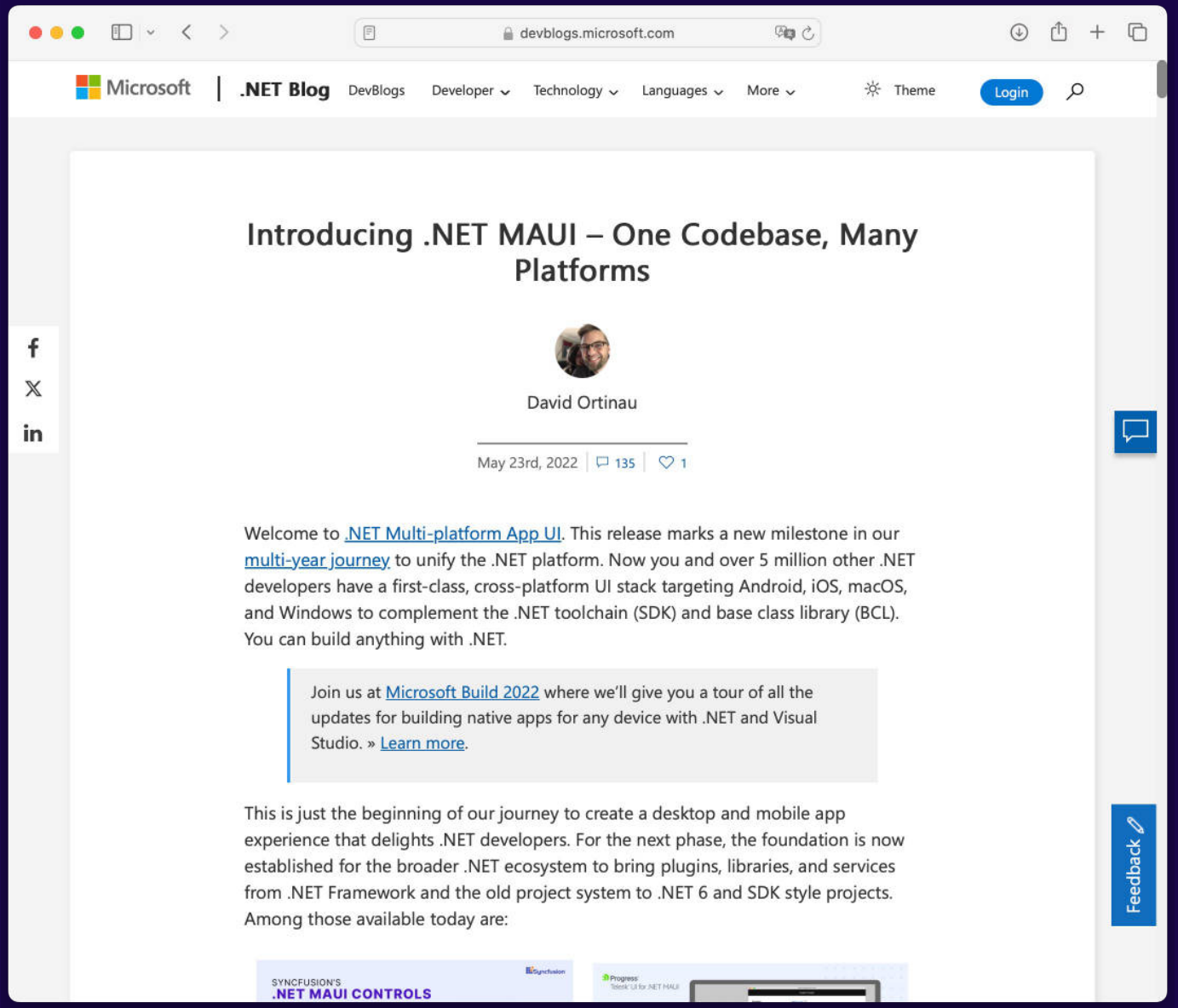


From .NET 6 to .NET 9



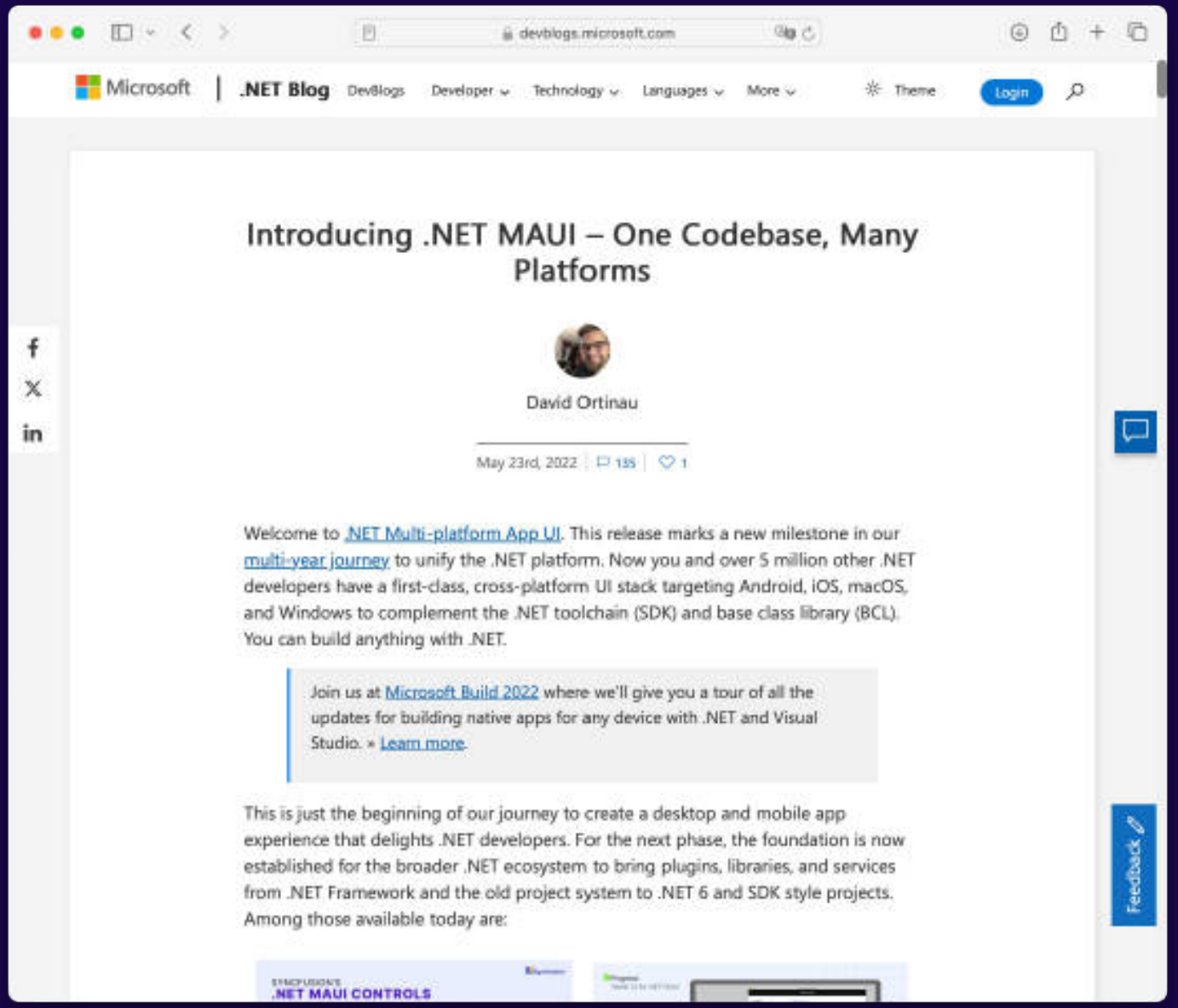
.NET 6

“This is just the beginning of our journey to create a desktop and mobile app experience that delights .NET developers. For the next phase, the foundation is now established for the broader .NET ecosystem to bring plugins, libraries, and services from .NET Framework and the old project system to .NET 6 and SDK style projects.”



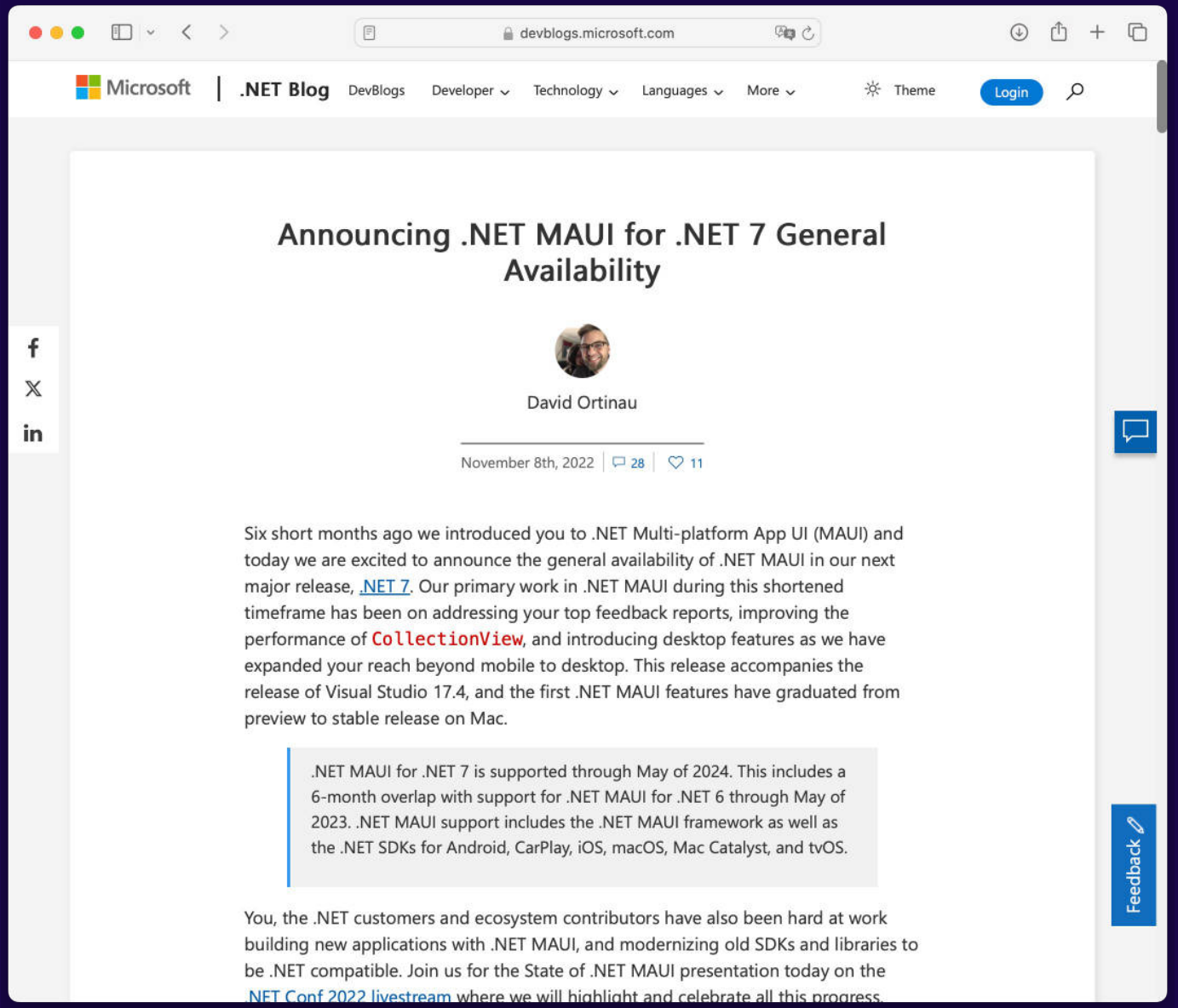
.NET 6

“This is just the beginning of our journey to create a desktop and mobile app experience that delights .NET developers. For the next phase, the foundation is now established for the broader .NET ecosystem to bring plugins, libraries, and services from .NET Framework and the old project system to .NET 6 and SDK style projects.”



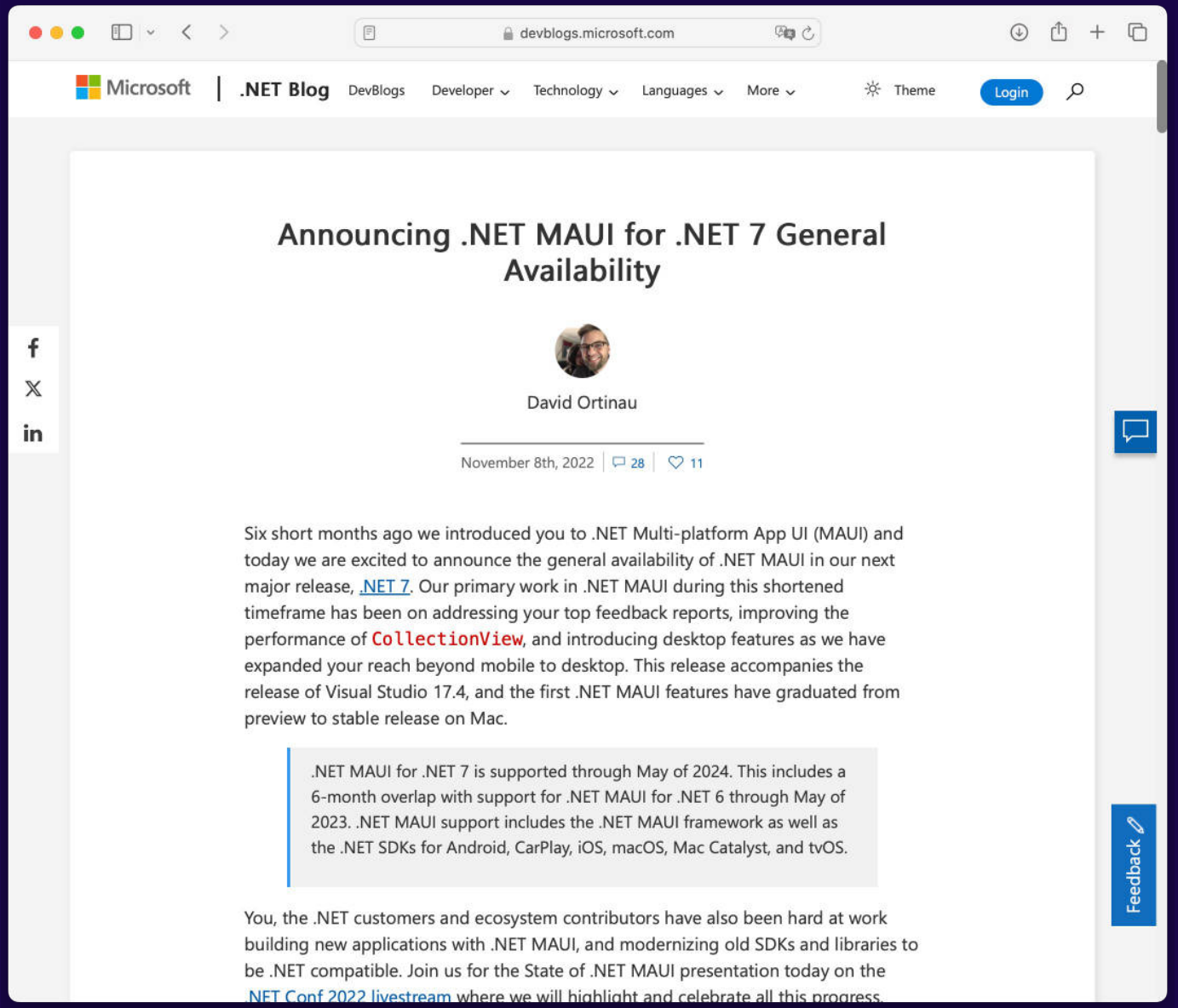
.NET 7

“Our primary work in .NET MAUI during this shortened timeframe has been on addressing your top feedback reports, improving the performance of CollectionView, and introducing desktop features as we have expanded your reach beyond mobile to desktop. This release accompanies the release of Visual Studio 17.4, and the first .NET MAUI features have graduated from preview to stable release on Mac.”

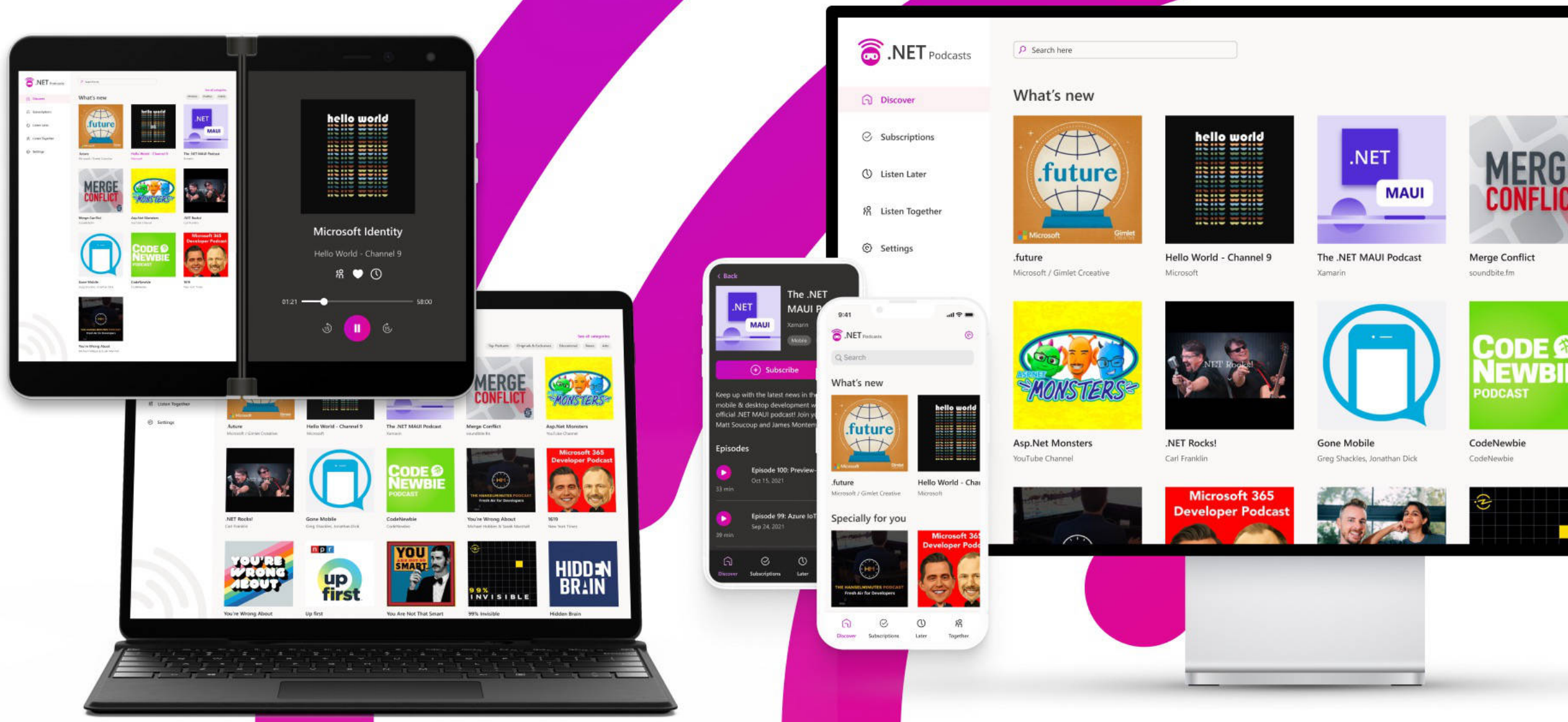


.NET 7

“Our primary work in .NET MAUI during this shortened timeframe has been on addressing your top feedback reports, improving the performance of CollectionView, and introducing desktop features as we have expanded your reach beyond mobile to desktop. This release accompanies the release of Visual Studio 17.4, and the first .NET MAUI features have graduated from preview to stable release on Mac.”

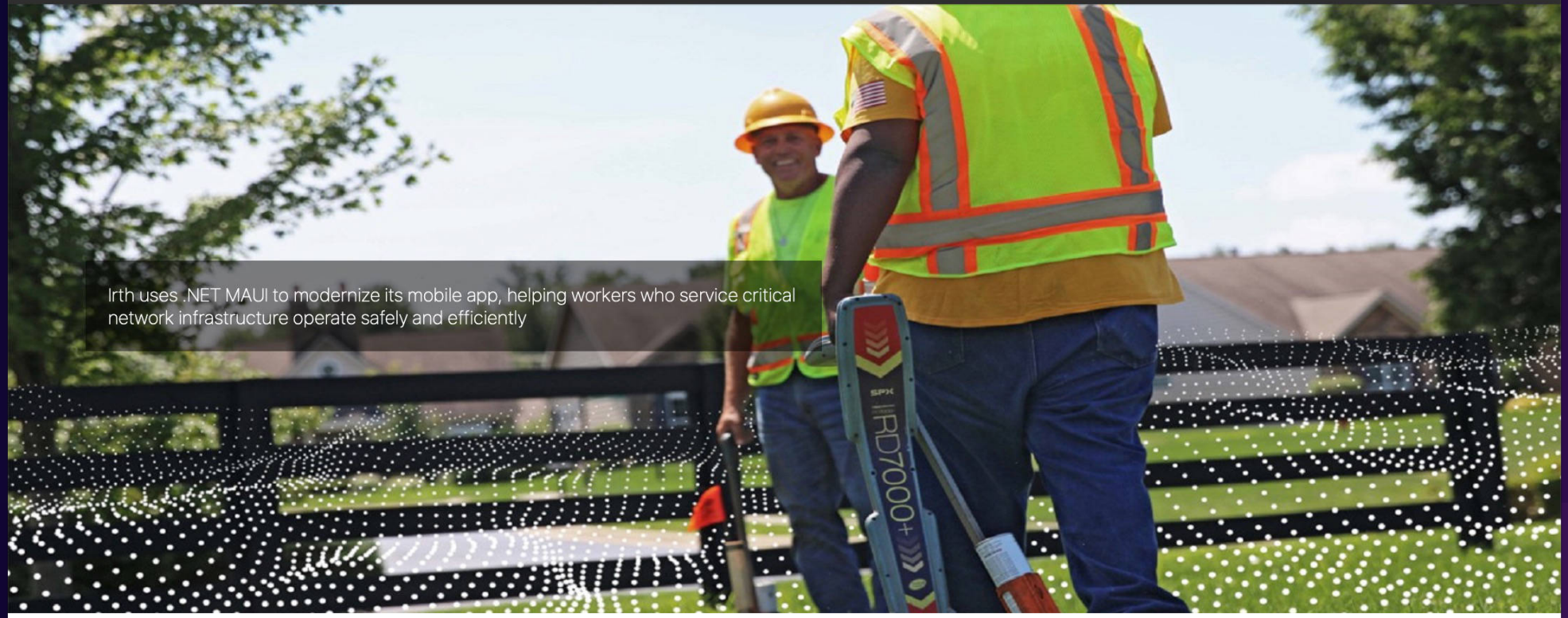


.NET Podcasts



ms-f1-sites-03-ea.azurewebsites.net

Microsoft | Microsoft Security | Azure | Dynamics 365 | Microsoft 365 | Microsoft Teams | Windows 365 | All Microsoft



Irth uses .NET MAUI to modernize its mobile app, helping workers who service critical network infrastructure operate safely and efficiently

May 31, 2023

Print

Technology company Irth Solutions (Irth) rebuilt its risk management, damage

Learn More

.NET MAUI

Azure Cloud Services


9:53
95%
Irth


Retail Insight


hogeschool
vives

irth
UtiliSphere

S


Xari

OE
XA

MAHIZ
grow your business

H3Y

ups

CARDIFF
CAERDYDD


Microsoft
Dynamics 365

Microsoft
Azure

CIVICA

SPORTS
NEXT
SPORTS×TECH

DevInMotion

Belgian
Red Cross
Flanders

ambient

A

insight

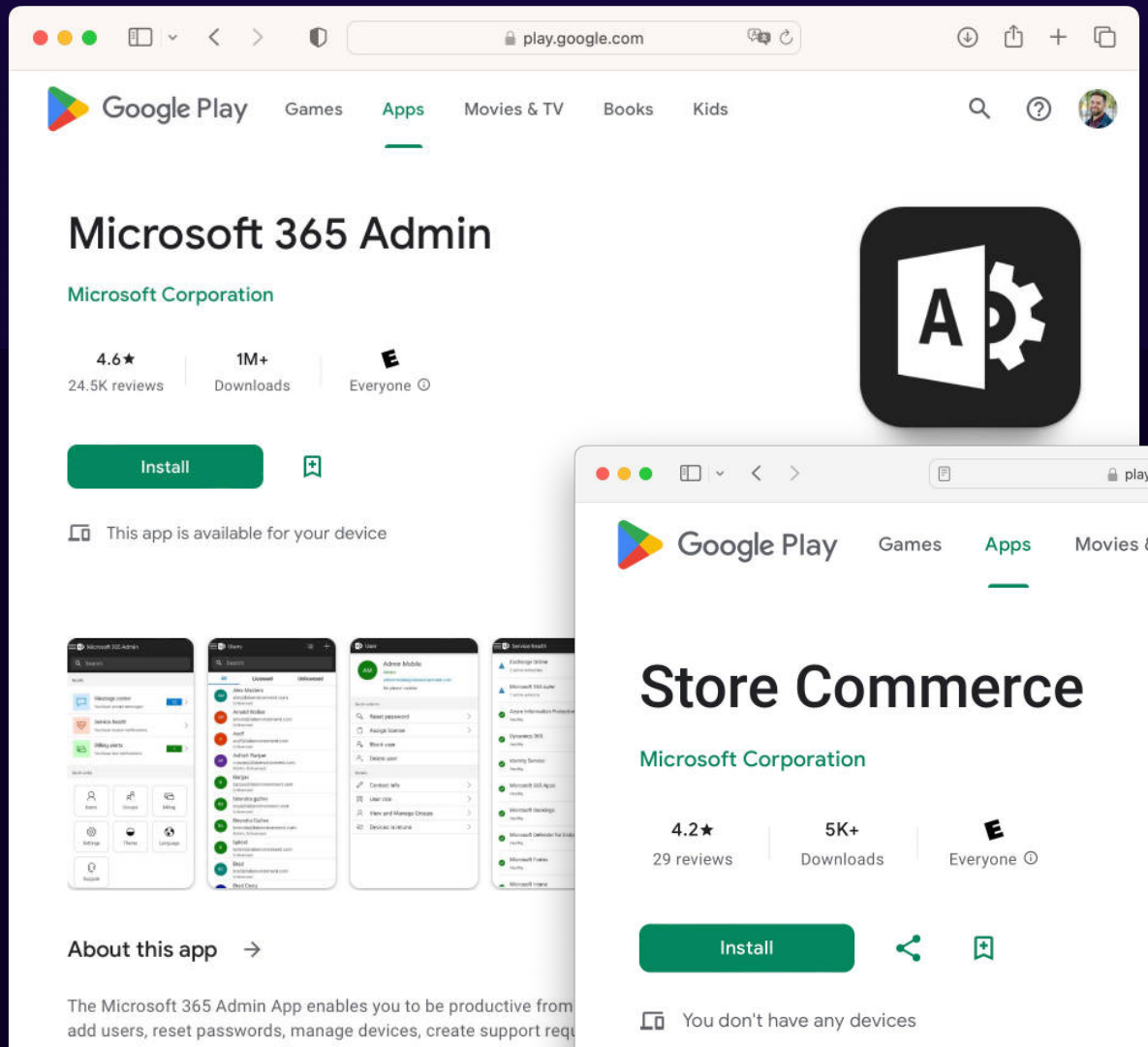
esri

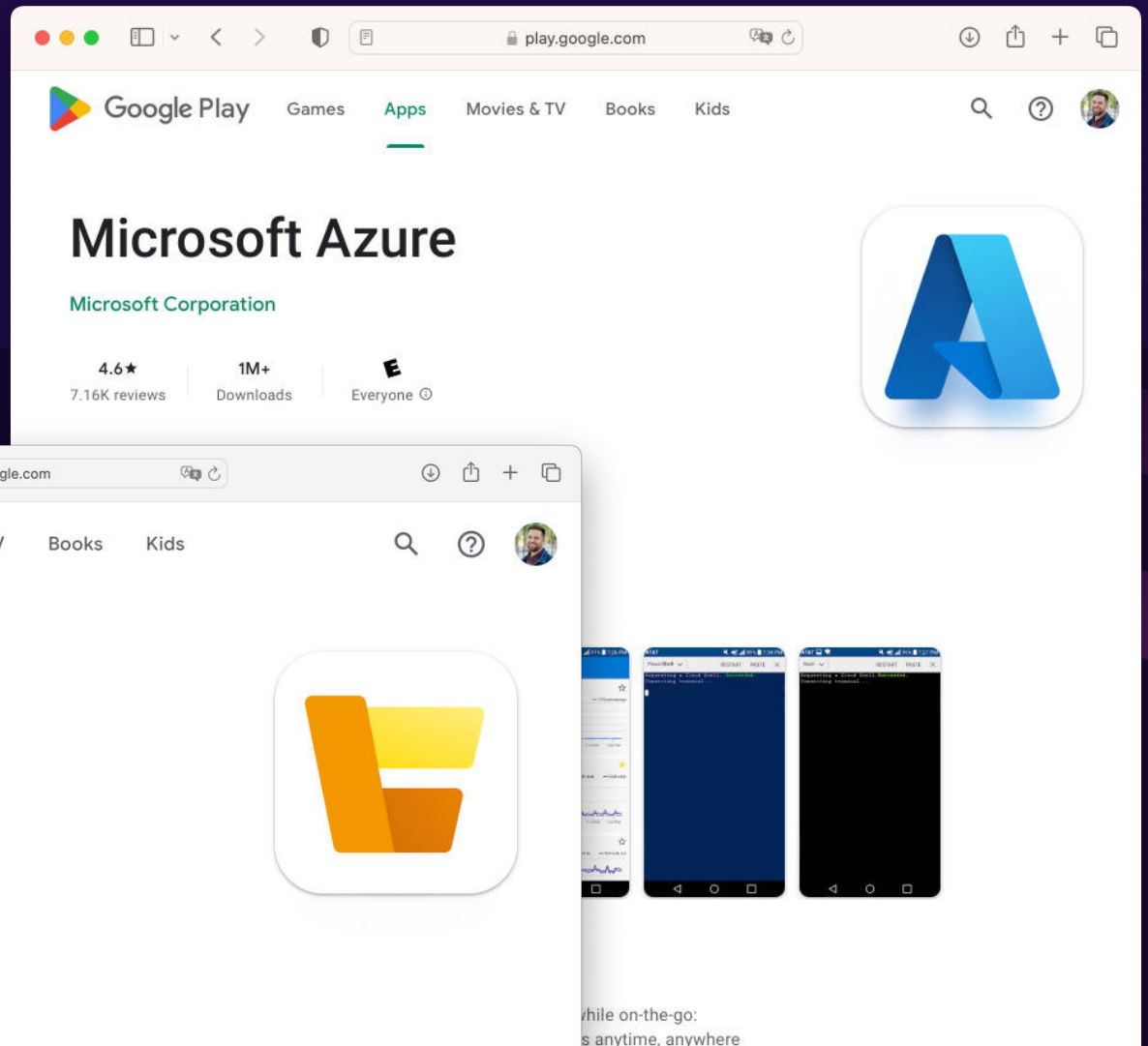

Soluzione 1

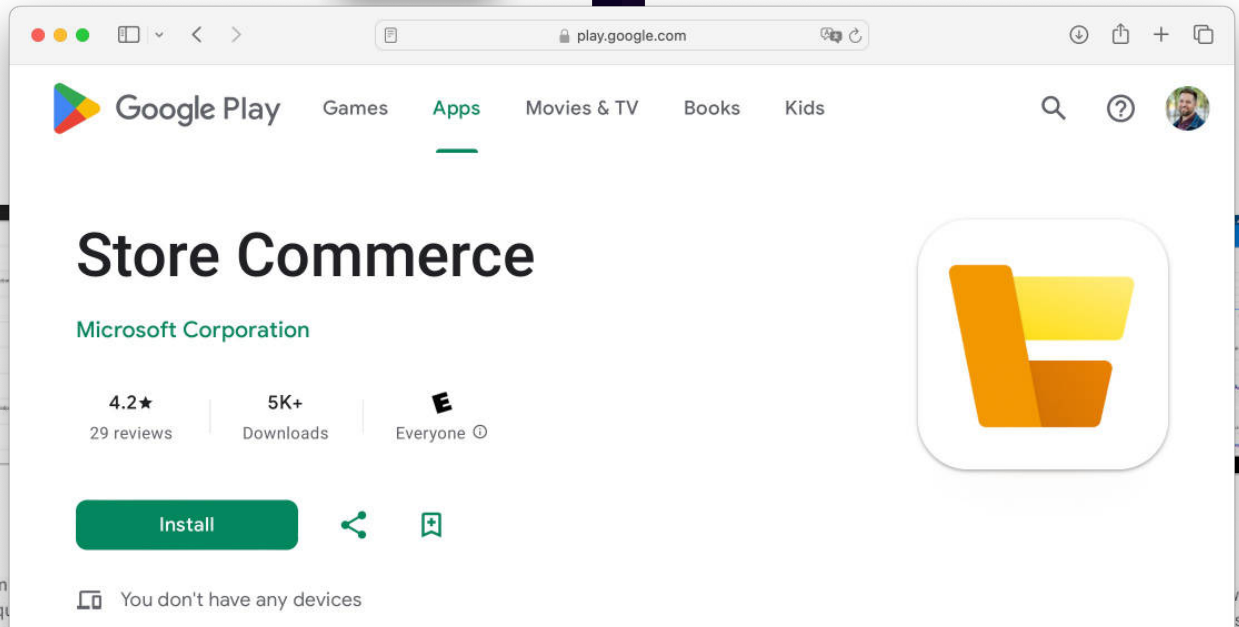
REY
TECHNOLOGY

[A]


.NET MAUI at Microsoft

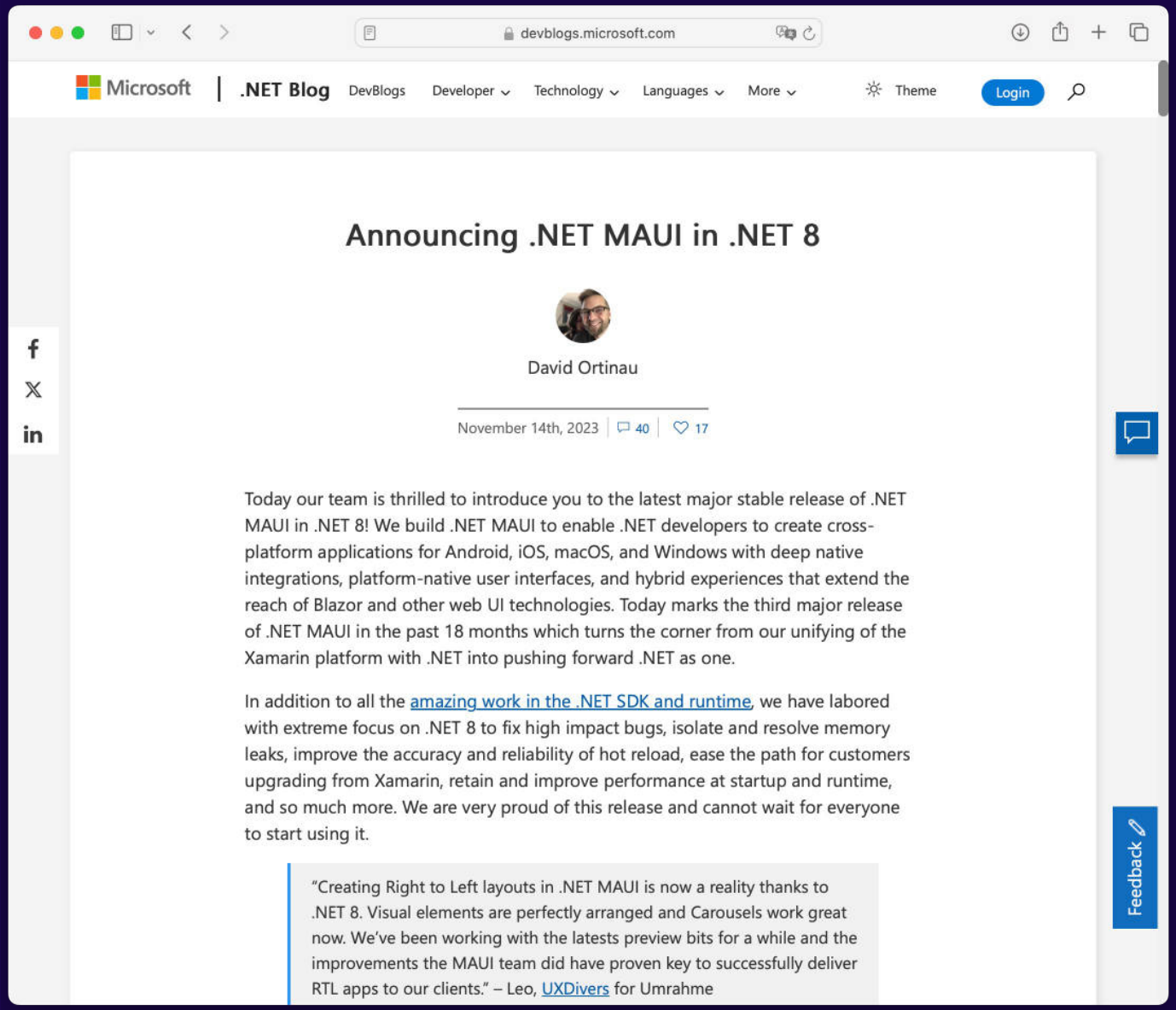






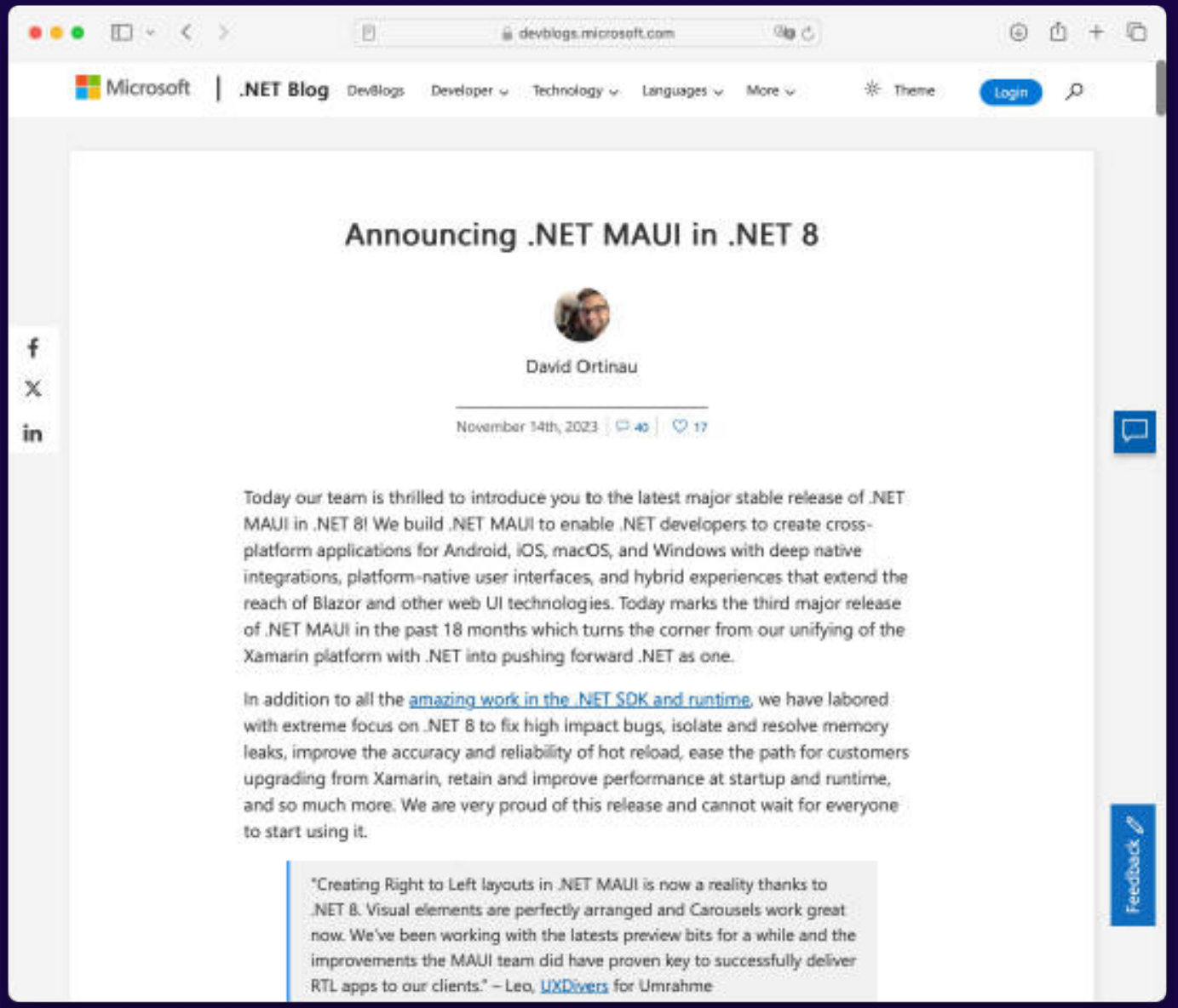
.NET 8

“Compared to the .NET 7 GA release that’s a 180% increase in pull requests merged and 283% more bug issues resolved. Since .NET 7 was a shorter release for .NET MAUI, I thought it would be interesting to include all the .NET 7 service releases as well. Taking that into account, .NET 8 still has 18% more pull requests merged and 13% more bug issues resolved.”



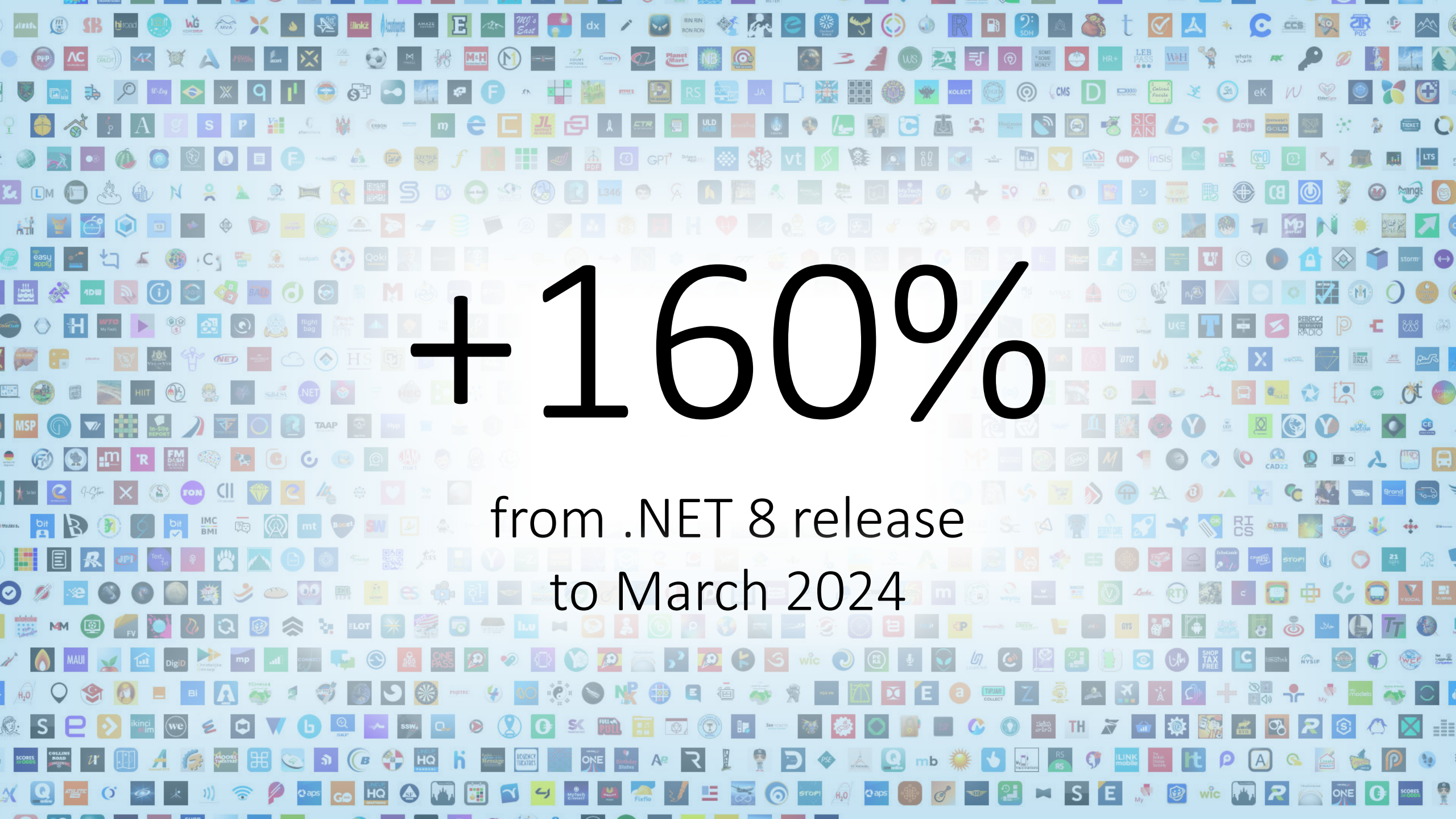
.NET 8

“Compared to the .NET 7 GA release that’s a 180% increase in pull requests merged and 283% more bug issues resolved. Since .NET 7 was a shorter release for .NET MAUI, I thought it would be interesting to include all the .NET 7 service releases as well. Taking that into account, .NET 8 still has 18% more pull requests merged and 13% more bug issues resolved.”




Google Play


Big 10 Mart
DFU
ES
TwinCAT
EchoGuide
CINEMA 1 PLUS
STOP!
21 ngán
NaCCA
Azure Fundamentals AZ-900 Exam
E-ZPass TOLL PLATE
CWEP
m
weyou App
Wander 120
Locke
RTS
V SOCIAL
MyWHA
GP BMT
M4M
FV
THE LOT
2022
GIDEON NOW
Mainland Football
BAJit
Bengaluru Kabaddi
GREEN PRESCRIPTION
APA
GYS
BRIDGE
COCO PLAY
حلال
TT
kamstrup
PRO VIKINGS
MAUI
DigiD
Christelijk Omroep
mp
CONNECT
585 RETAIL
ONE PASS
TRACKMAN
wic
PEKA
limolink
NYSIF
IWCF
Net Logger Companion
H2O
Bi
YGO VN
FUJITEC
TiPJAR COLLECT
SKF
SSW
My
afp modelo
BUCKEYE
ikinci elim
we
SCORES AND ODDS
COLLINS ROAD THEATRES
Ascend EVV
MOORE THEATRES
LINEUP HQ FANDUEL
Radio Message
REGENCY THEATRES
METRO ONE
Birthday Status
PSE
ST. MICHAEL
QSR online
mb
My Vaccinations
RS
The British Horse Society
Create CRAFT
Coyote
ATHLETIC
aps
GO BMD

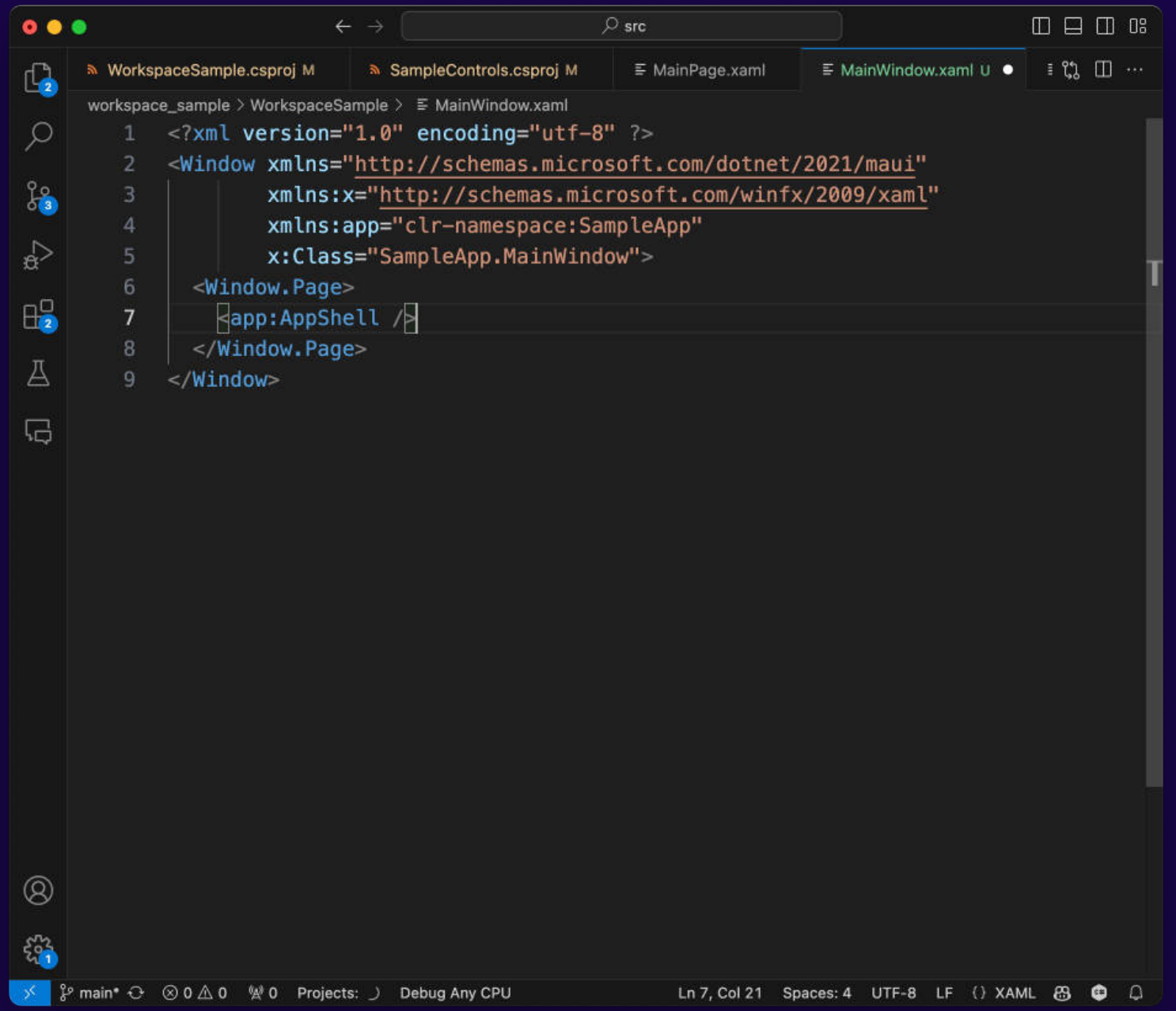

+160%
from .NET 8 release
to March 2024

DEMO

New in .NET 9

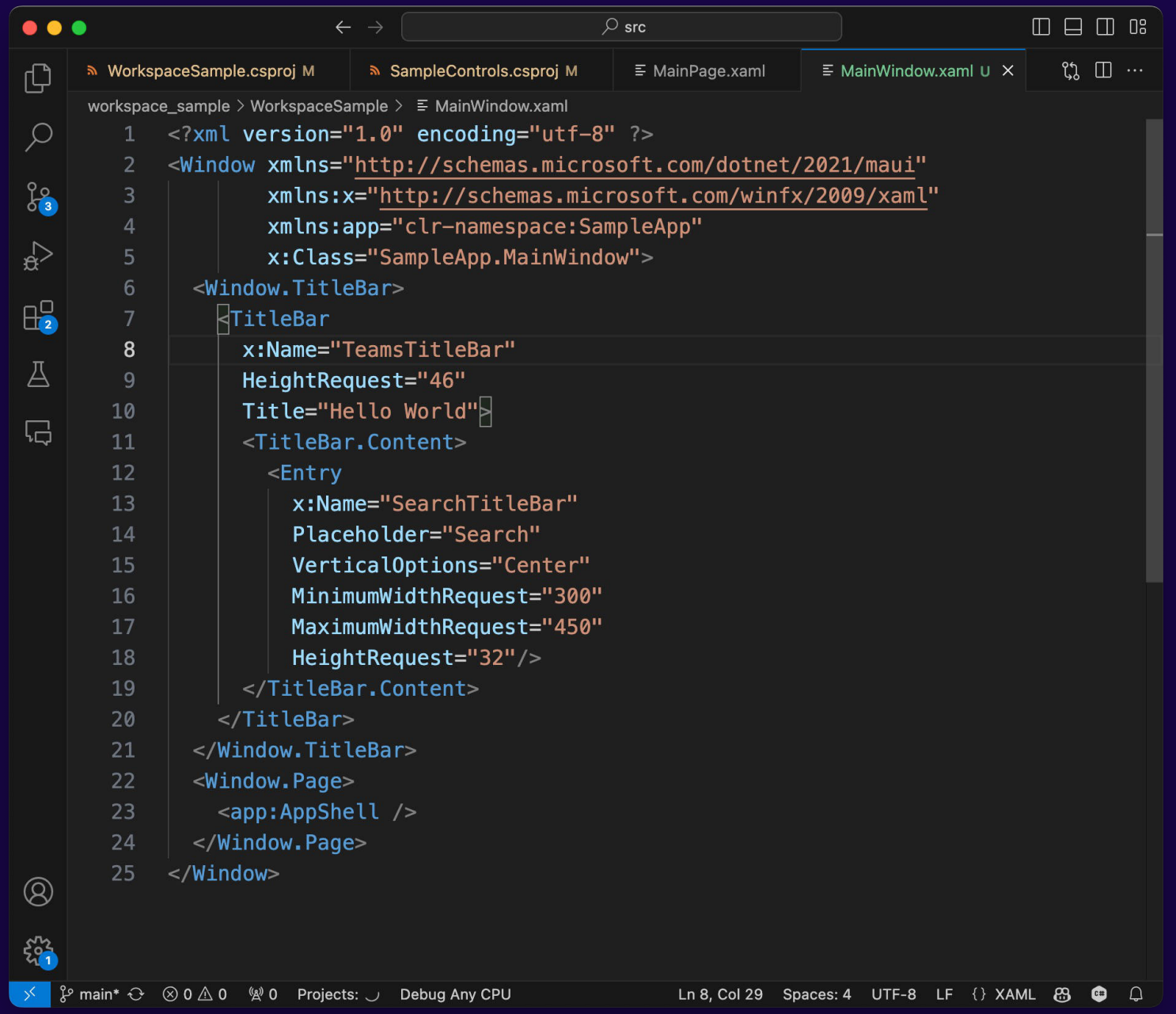
MainWindow.xaml

- Window already exists
- MainWindow just reveals it by default



TitleBar

- Windows-only (right now)
- PR #23019

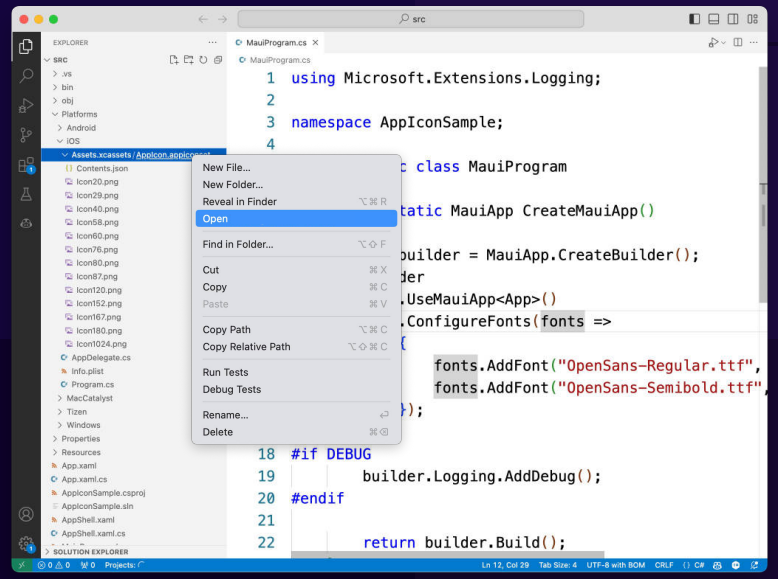


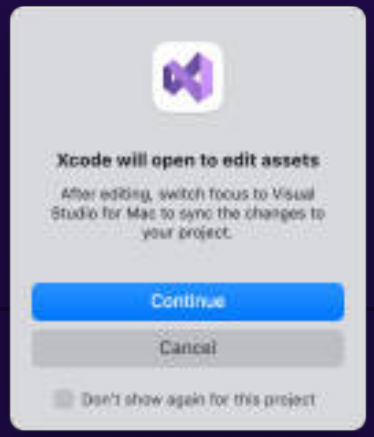

TitleBar
Content Options
Color Options

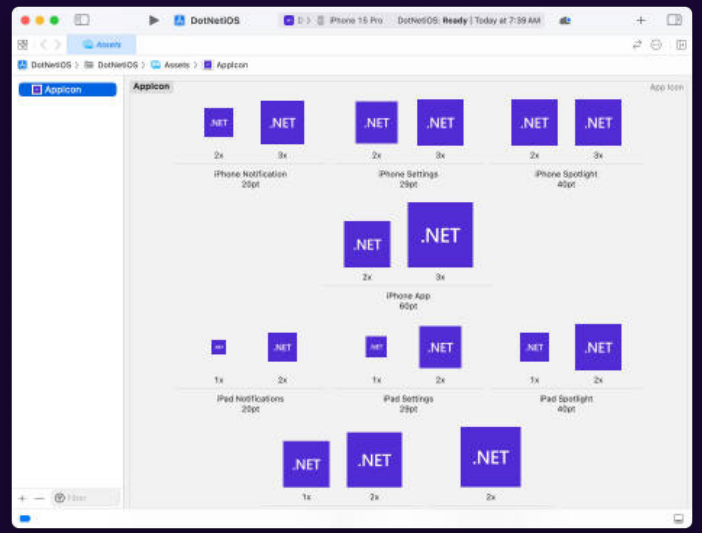
Xcode Sync

- Xcassets
- Storyboards
- Xibs

Utilize command line

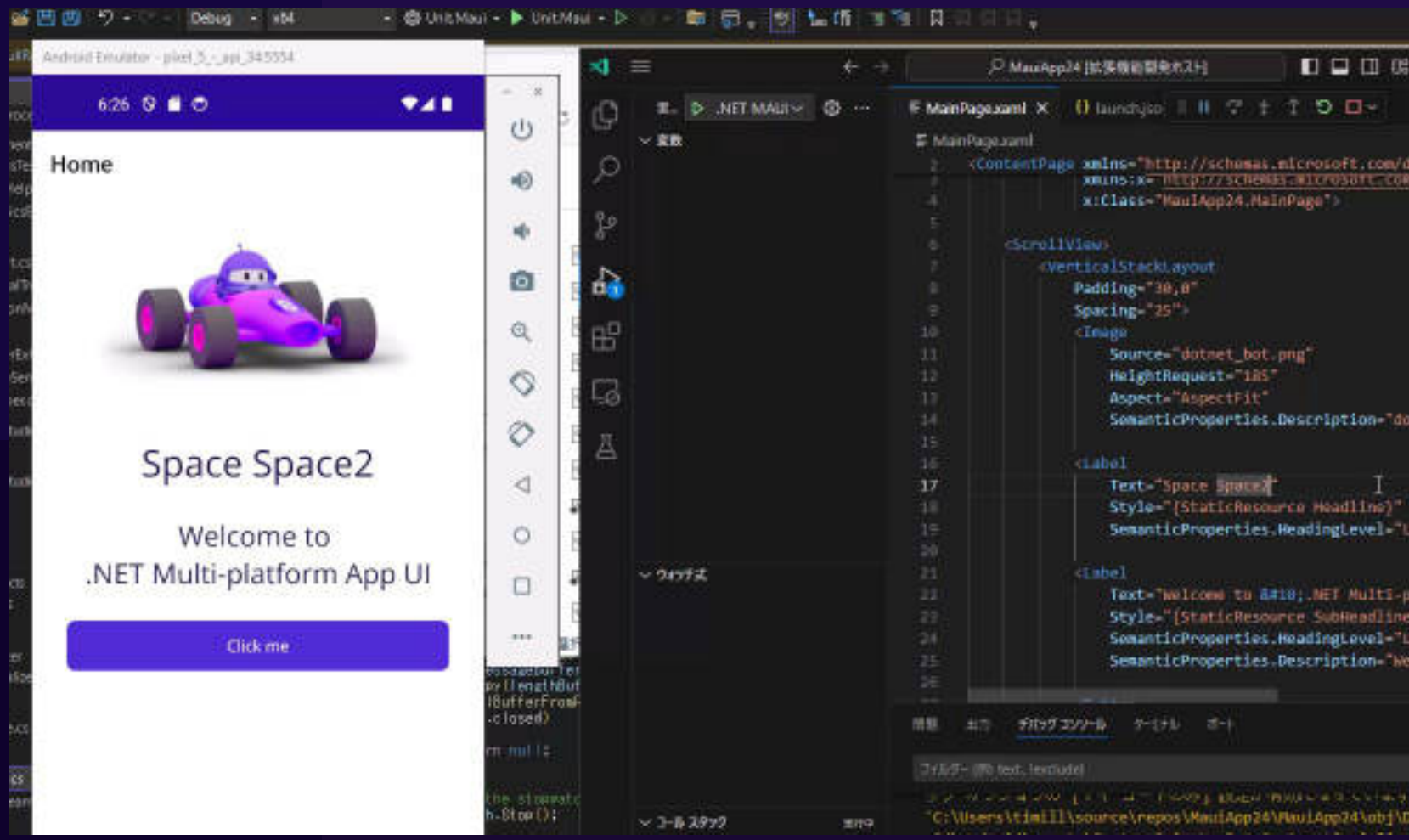






MAUI DevKit

- Core Debugger
- XAML Intellisense
- XAML and C# Hot Reload
  - Android
  - iOS
  - Mac Catalyst
  - Windows



iOS Library Multi-targeting

Libraries can target multiple distinct iOS versions in the same project.

To produce two libraries, one for iOS 17.0 bindings, and one for iOS 17.2 bindings:

```
<TargetFrameworks>net9.0-ios17.0;net9.0-ios17.2</TargetFrameworks>
```

Android Asset Packs

Places assets in a separate package (asset pack) to reduce your app size.

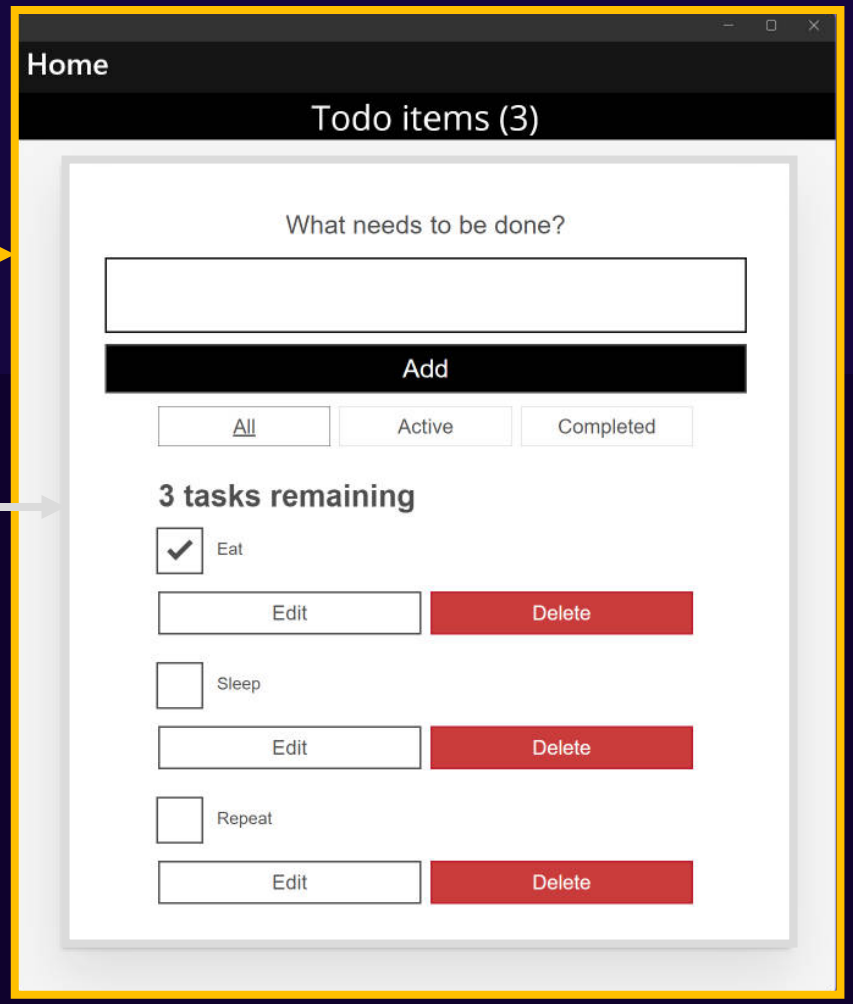
Types: InstallTime (default), FastFollow, OnDemand

```
<MauiAsset
    Update="Resources\Raw\myvideo.mp4"
    AssetPack="myassetpack"
    DeliveryType="FastFollow" />
```

.NET MAUI HybridWebView

Native app container & controls

Hybrid WebView React JS



Multi-platform control based on .NET MAUI WebView

Host static web assets (HTML, JS, CSS, etc.)

Communicate between JavaScript & .NET

Reuse your Angular, React, etc. investments in MAUI native desktop and mobile apps

Use any web-based UI and it can interact with C# code

```
<PropertyGroup Condition="$(TargetFramework.Contains('-
maccatalyst') or $(TargetFramework.Contains('-ios')">
    <PublishAot>true</PublishAot>
    <EnableTrimAnalyzer>true</EnableTrimAnalyzer>
    <EnableAotAnalyzer>true</EnableAotAnalyzer>
    <EnableSingleFileAnalyzer>true</EnableSingleFileAnalyzer>
</PropertyGroup>
```

Native AOT

Ahead of time compilation that results in smaller and faster apps.

All code must be trimmable.

```
dotnet publish -f net9.0-maccatalyst -r maccatalyst-arm64
```

	Metric	Mono	NativeAOT	Diff
1	Package size	13,850,984 B	5,301,713 B	-8,549,271 B (-61.7%)
2	Size on disk	40,051,550 B	12,673,754 B	-27,377,796 B (-68.4%)
3	Startup time	209.6 ms	108.3 ms	-101.3 ms (-48.3%)

Native AOT

Ahead of time compilation that results in smaller and faster apps.

All code must be trimmable.

Interop with Slim Bindings

What is it?

- A slim binding is a lightweight approach to interop between .NET and Obj-C/Swift/Kotlin/Java.
- **Why?**
- Binding directly to a library is a complicated process that has exhausted the patience of many respectable developers.

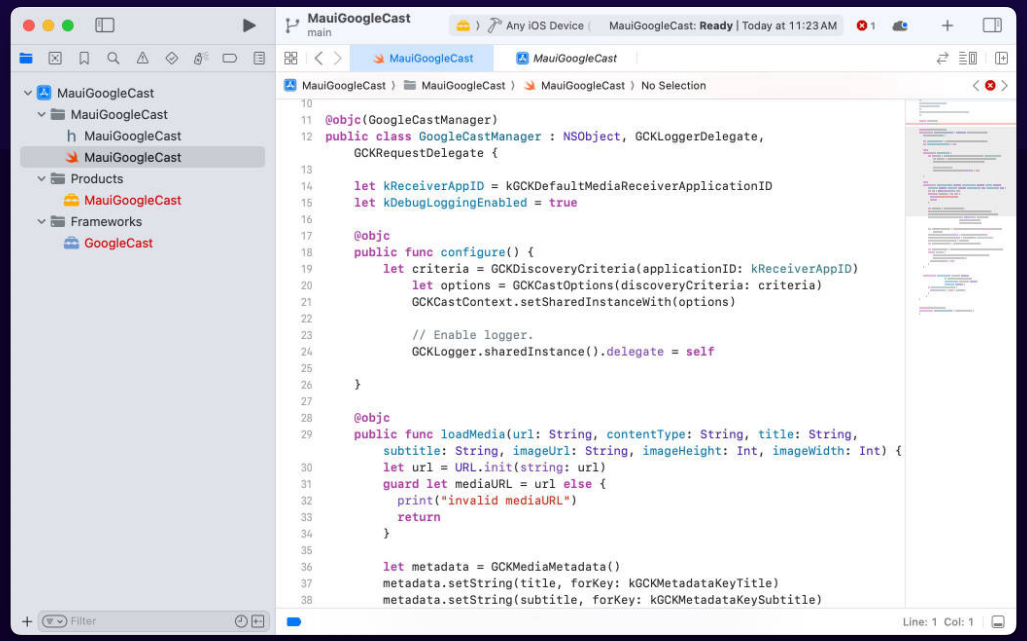
How do you make one? You will need:

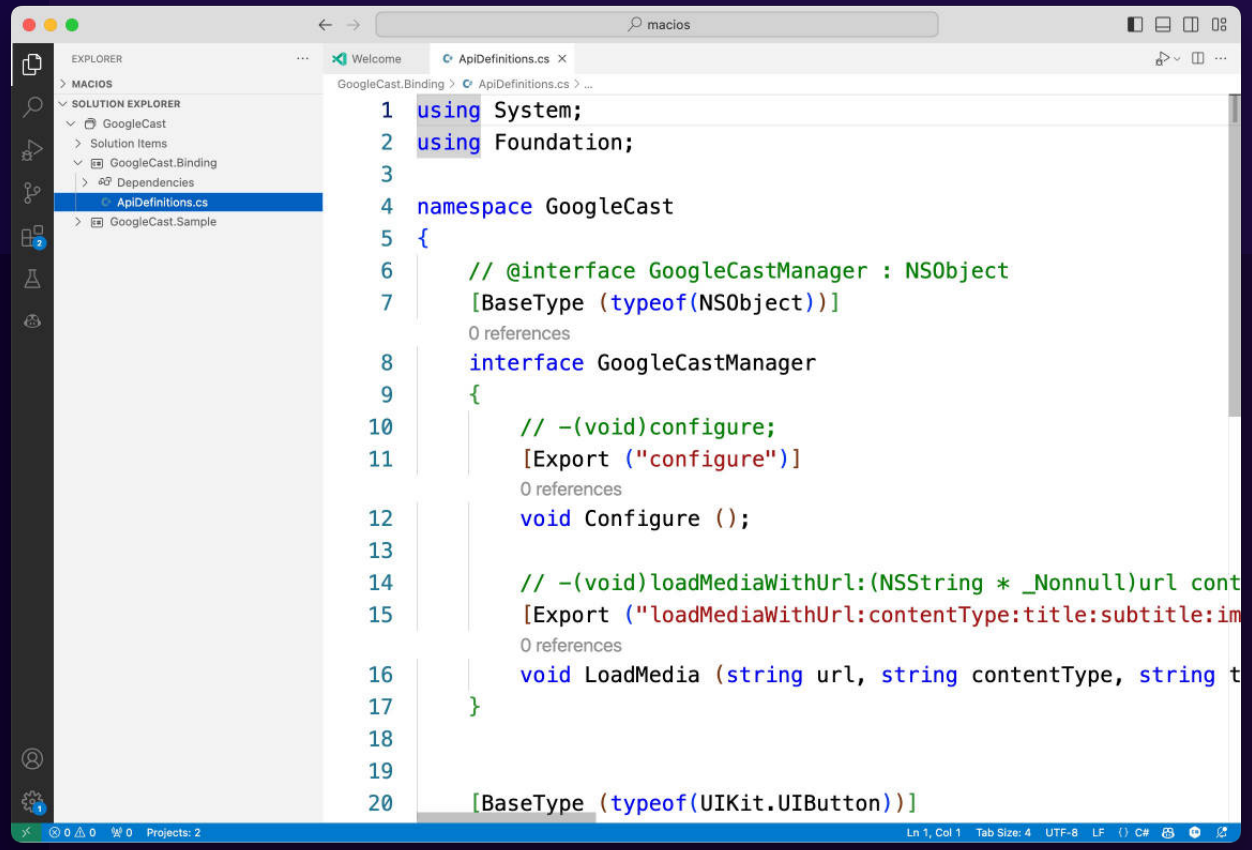
- Xcode and/or Android project that implements the library following vendor docs
- A .NET class library that interops with your limited API
- (optional) Sample .NET MAUI app
- Build a NuGet

https://aka.ms/maui/slim-bindings

Interop with Slim Bindings

In Xcode define and implement the methods you need. Mirror this minimal surface in .NET.





Slim Binding: Use in your app

```
if (googleCastManager is null)
{
    googleCastManager = new
GoogleCast.GoogleCastManager();
    googleCastManager.Configure();
}

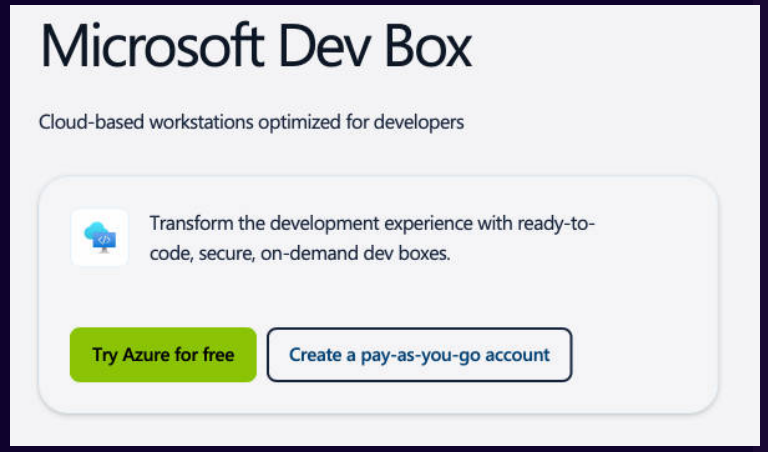

googleCastManager.LoadMedia("https://www.sample-
videos.com/video123/mp4/720/big_buck_bunny_720p_1mb.mp4",
"video/mp4","Big Buck Bunny", "Big Buck Bunny (open-source
movie)","https://peach.blender.org/wp-
content/uploads/title_anouncement.jpg",100,100);
```

RETIREMENT

Visual Studio for Mac

- 17.6 is the final release
- Supported through August 31, 2024

Options:





https://devblogs.microsoft.com/visualstudio/visual-studio-for-mac-retirement-announcement/

RETIREMENT

Visual Studio App Center

Supported through March 31, 2025

App Center capability	Recommended alternative solution
Build	Azure Pipelines
Test	BrowserStack App Automate
Distribution	Apple's AppStore for iOS app production releases and Apple's TestFlight for iOS app test releases. Google Play for production releases and Google Play Console for test releases. Azure Pipelines tasks can be used for **distributing** to AppStore/TestFlight and Google Play. For Microsoft Store applications, we recommend using Package Flights feature available in Partner Center.
Analytics & diagnostics	Datadog Dynatrace New Relic

https://learn.microsoft.com/appcenter/retirement

Component Vendors











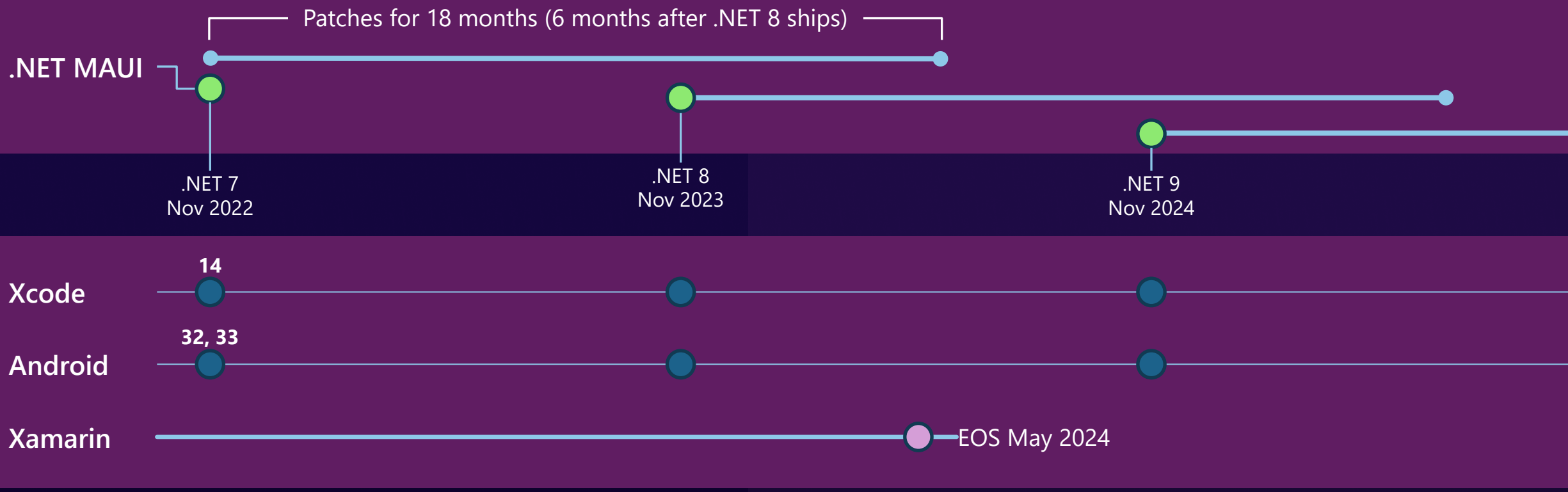








Release schedule, support & patching

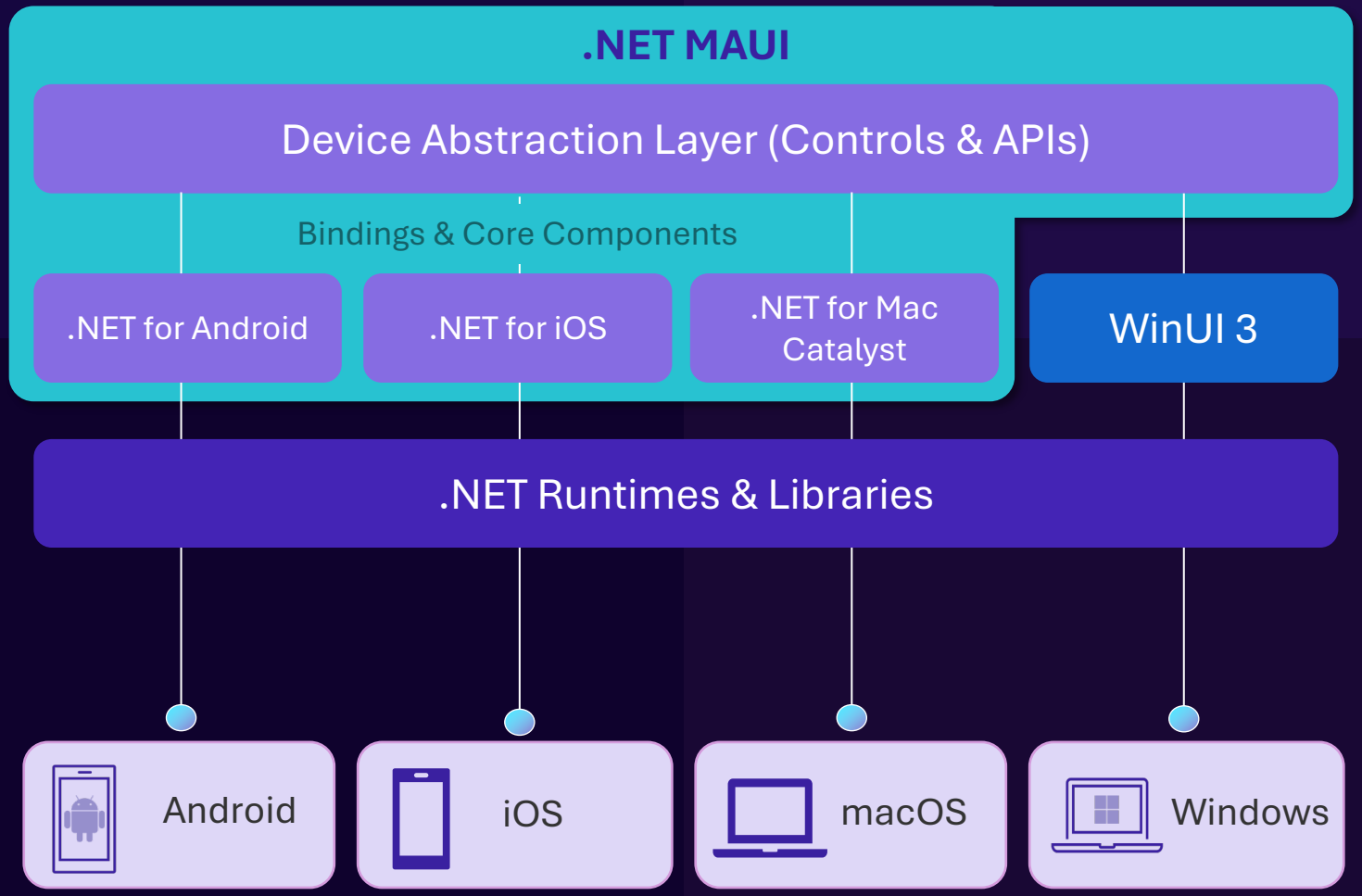


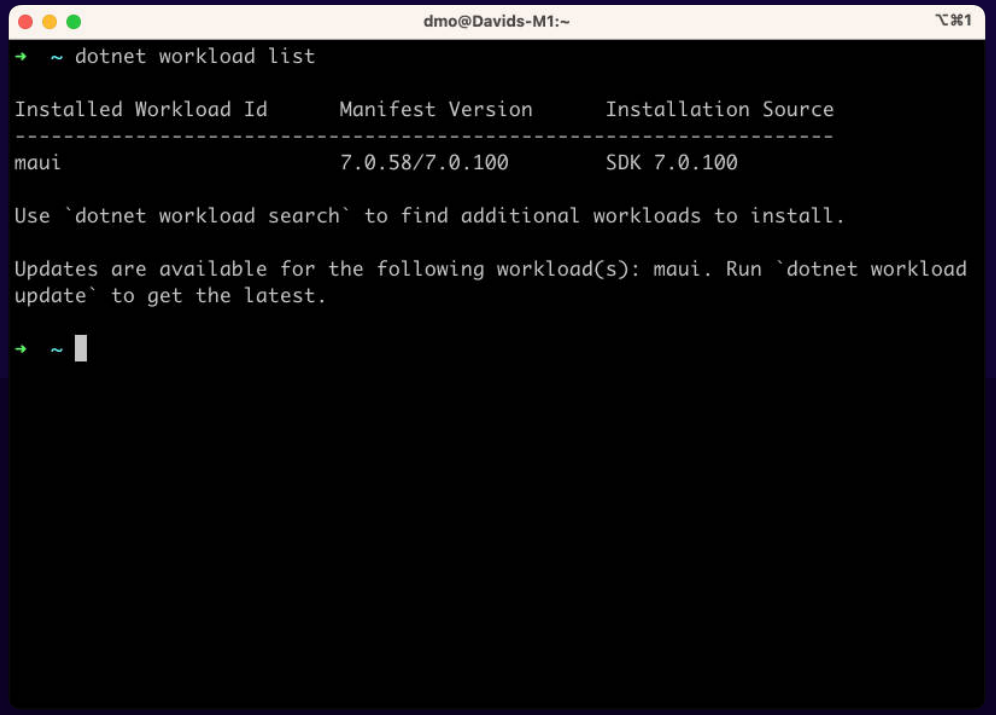
- A major version of .NET MAUI will ship with .NET going forward (.NET 7, .NET 8, etc.)
- .NET MAUI is an optional workload with external dependencies (such as Xcode & Android SDK tools)
- A major version of .NET MAUI will receive patches for 6 months after a successor (major version) ships
- Xamarin SDKs will receive support until May 2024

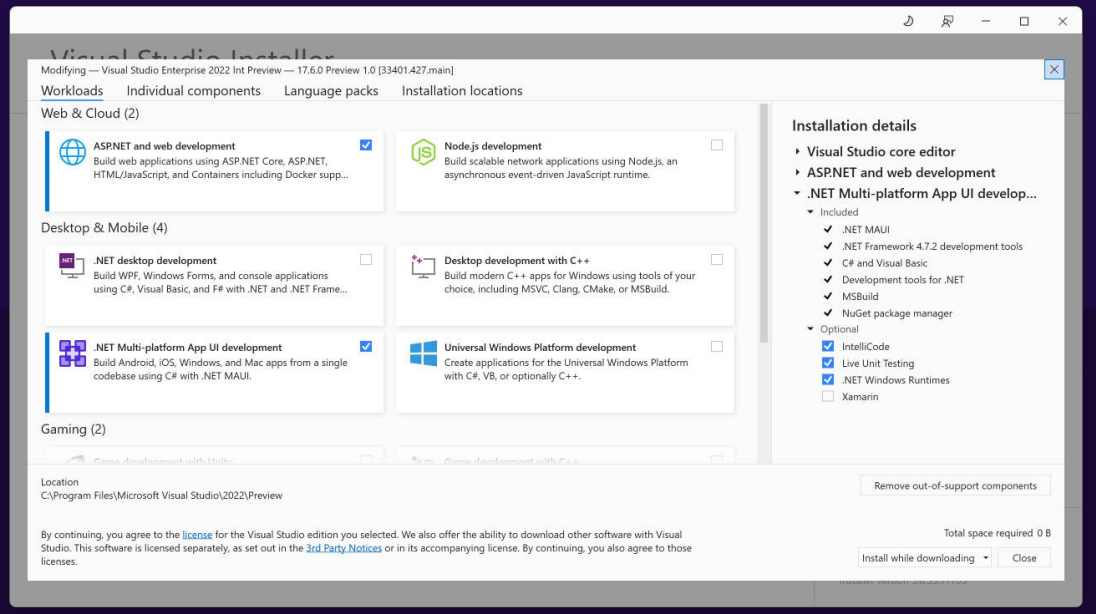
Thanks! Let’s talk.

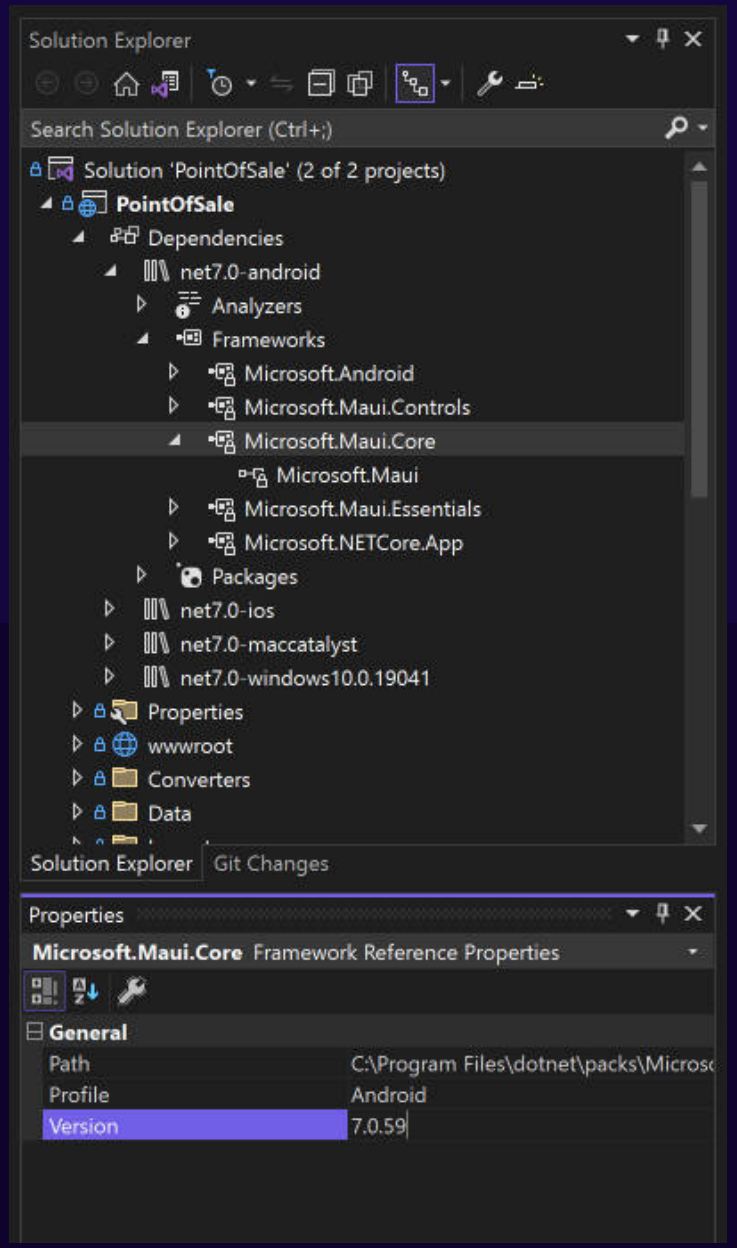
@davidortinau

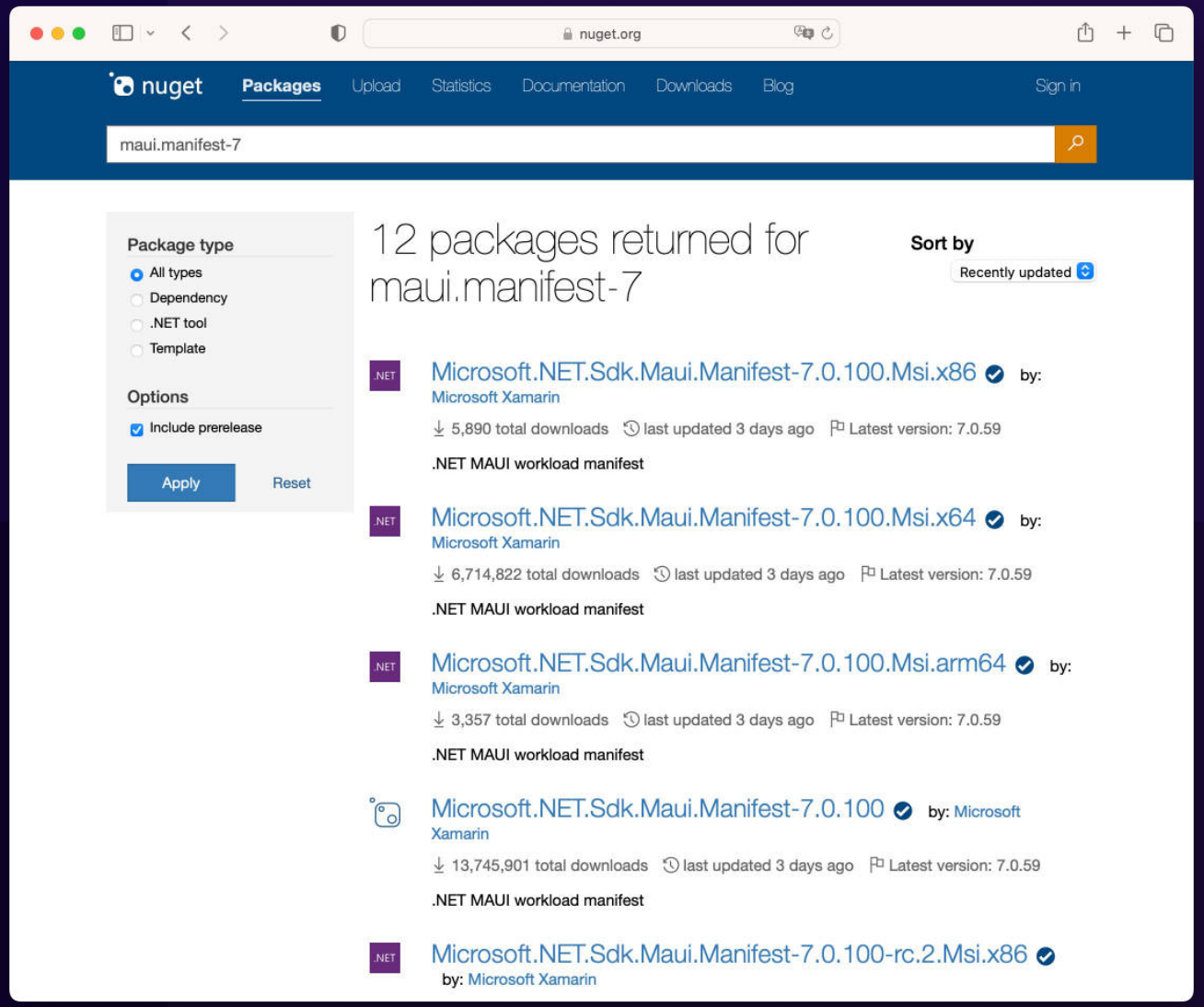
.NET Multi-platform App Components




dmo@Davids-M1:~
⌥⌘1
➜ ~ dotnet workload list
Installed Workload Id Manifest Version Installation Source
maui 7.0.58/7.0.100 SDK 7.0.100
Use `dotnet workload search` to find additional workloads to install.
Updates are available for the following workload(s): maui. Run `dotnet workload update` to get the latest.
➜ ~


Modifying — Visual Studio Enterprise 2022 Int Preview — 17.6.0 Preview 1.0 [33401.427.main]
Workloads
Individual components
Language packs
Installation locations
Web & Cloud (2)
ASP.NET and web development
Build web applications using ASP.NET Core, ASP.NET, HTML/JavaScript, and Containers including Docker supp...
Node.js development
Build scalable network applications using Node.js, an asynchronous event-driven JavaScript runtime.
Desktop & Mobile (4)
.NET desktop development
Build WPF, Windows Forms, and console applications using C#, Visual Basic, and F# with .NET and .NET Frame...
Desktop development with C++
Build modern C++ apps for Windows using tools of your choice, including MSVC, Clang, CMake, or MSBuild.
.NET Multi-platform App UI development
Build Android, iOS, Windows, and Mac apps from a single codebase using C# with .NET MAUI.
Universal Windows Platform development
Create applications for the Universal Windows Platform with C#, VB, or optionally C++.
Gaming (2)
Installation details
Visual Studio core editor
ASP.NET and web development
.NET Multi-platform App UI develop...
Included
.NET MAUI
.NET Framework 4.7.2 development tools
C# and Visual Basic
Development tools for .NET
MSBuild
NuGet package manager
Optional
IntelliCode
Live Unit Testing
.NET Windows Runtimes
Xamarin
Location
C:\Program Files\Microsoft Visual Studio\2022\Preview
Remove out-of-support components
By continuing, you agree to the license for the Visual Studio edition you selected. We also offer the ability to download other software with Visual Studio. This software is licensed separately, as set out in the 3rd Party Notices or in its accompanying license. By continuing, you also agree to those licenses.
Total space required 0 B
Install while downloading
Close


Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'PointOfSale' (2 of 2 projects)
PointOfSale
Dependencies
net7.0-android
Analyzers
Frameworks
Microsoft.Android
Microsoft.Maui.Controls
Microsoft.Maui.Core
Microsoft.Maui
Microsoft.Maui.Essentials
Microsoft.NETCore.App
Packages
net7.0-ios
net7.0-maccatalyst
net7.0-windows10.0.19041
Properties
wwwroot
Converters
Data
Solution Explorer
Git Changes
Properties
Microsoft.Maui.Core Framework Reference Properties
General
Path
C:\Program Files\dotnet\packs\Microsc
Profile
Android
Version
7.0.59


nuget.org
nuget
Packages
Upload
Statistics
Documentation
Downloads
Blog
Sign in
maui.manifest-7
Package type
All types
Dependency
.NET tool
Template
Options
Include prerelease
Apply
Reset
12 packages returned for maui.manifest-7
Sort by
Recently updated
Microsoft.NET.Sdk.Maui.Manifest-7.0.100.Msi.x86 by: Microsoft Xamarin
5,890 total downloads last updated 3 days ago Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100.Msi.x64 by: Microsoft Xamarin
6,714,822 total downloads last updated 3 days ago Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100.Msi.arm64 by: Microsoft Xamarin
3,357 total downloads last updated 3 days ago Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100 by: Microsoft Xamarin
13,745,901 total downloads last updated 3 days ago Latest version: 7.0.59
.NET MAUI workload manifest
Microsoft.NET.Sdk.Maui.Manifest-7.0.100-rc.2.Msi.x86
by: Microsoft Xamarin

dotnet.microsoft.com

Download .NET 7.0

Not sure what to download? See recommended downloads for the latest version of .NET.

7.0.2

Release notes **Latest release date** January 10, 2023

Build apps - SDK

SDK 7.0.102

OS	Installers	Binaries
Linux	Package manager instructions	Arm32 | Arm32 Alpine | Arm64 | Arm64 Alpine | x64 | x64 Alpine
macOS	Arm64 | x64	Arm64 | x64
Windows	Arm64 | x64 | x86 | winget instructions	Arm64 | x64 | x86
All	dotnet-install scripts	

Visual Studio support
Visual Studio 2022 (v17.4)
Visual Studio 2022 for Mac (v17.4)

Included in
Visual Studio 17.4.4

Run apps - Runtime

ASP.NET Core Runtime 7.0.2

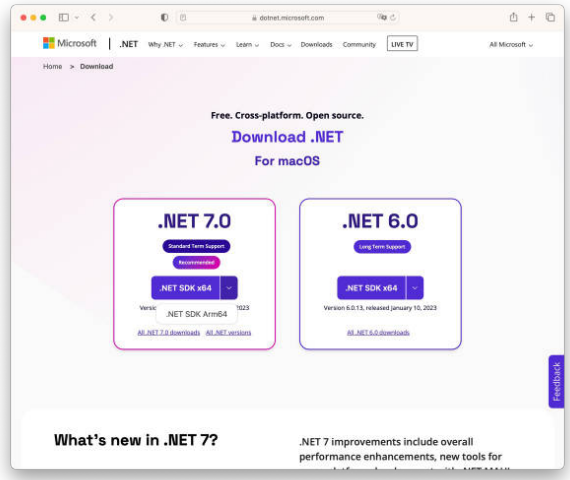
The ASP.NET Core Runtime enables you to run existing web/server applications. **On Windows, we recommend installing the Hosting Bundle, which includes the .NET Runtime and IIS support.**

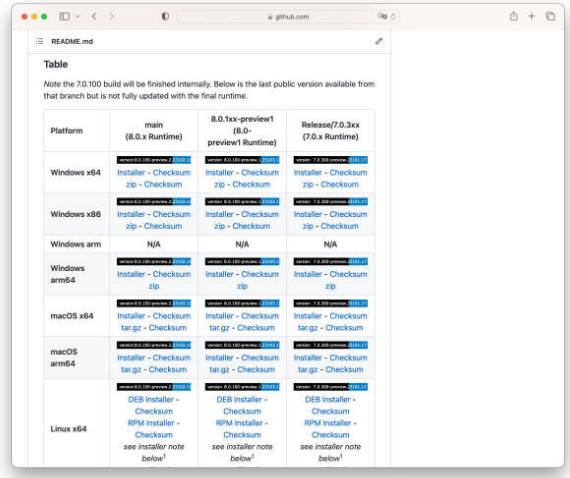
IIS runtime support (ASP.NET Core Module v2)
17.0.22341.2

OS	Installers	Binaries
Linux	Package manager instructions	Arm32 | Arm32 Alpine | Arm64 | Arm64 Alpine | x64 | x64 Alpine
macOS		Arm64 | x64
Windows	Hosting Bundle | x64 | x86 | winget instructions	Arm64 | x64 | x86

.NET Desktop Runtime 7.0.2

Feedback





Commands

> dotnet sdk check

> dotnet –version

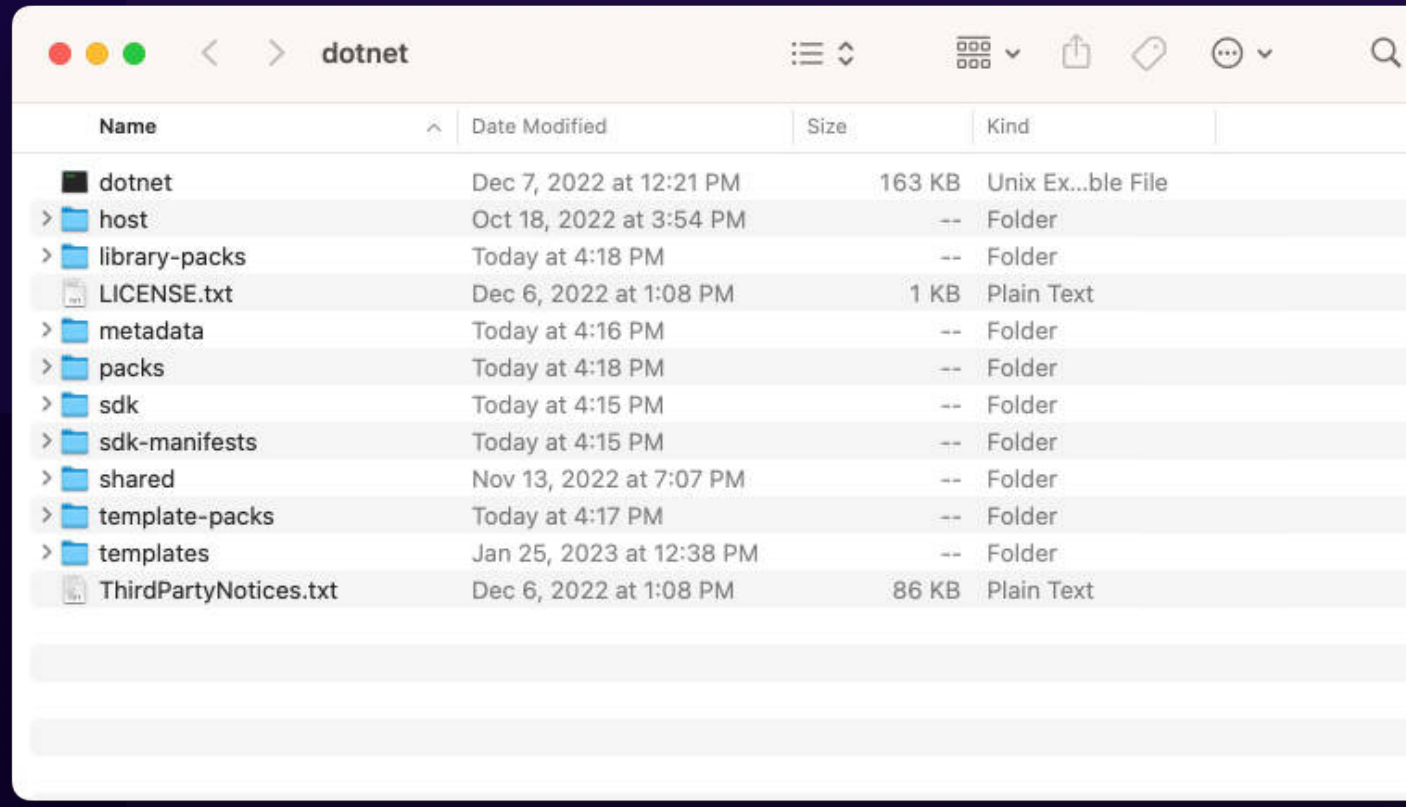
> dotnet workload list

> dotnet workload install maui (sudo)

> dotnet workload update (sudo)

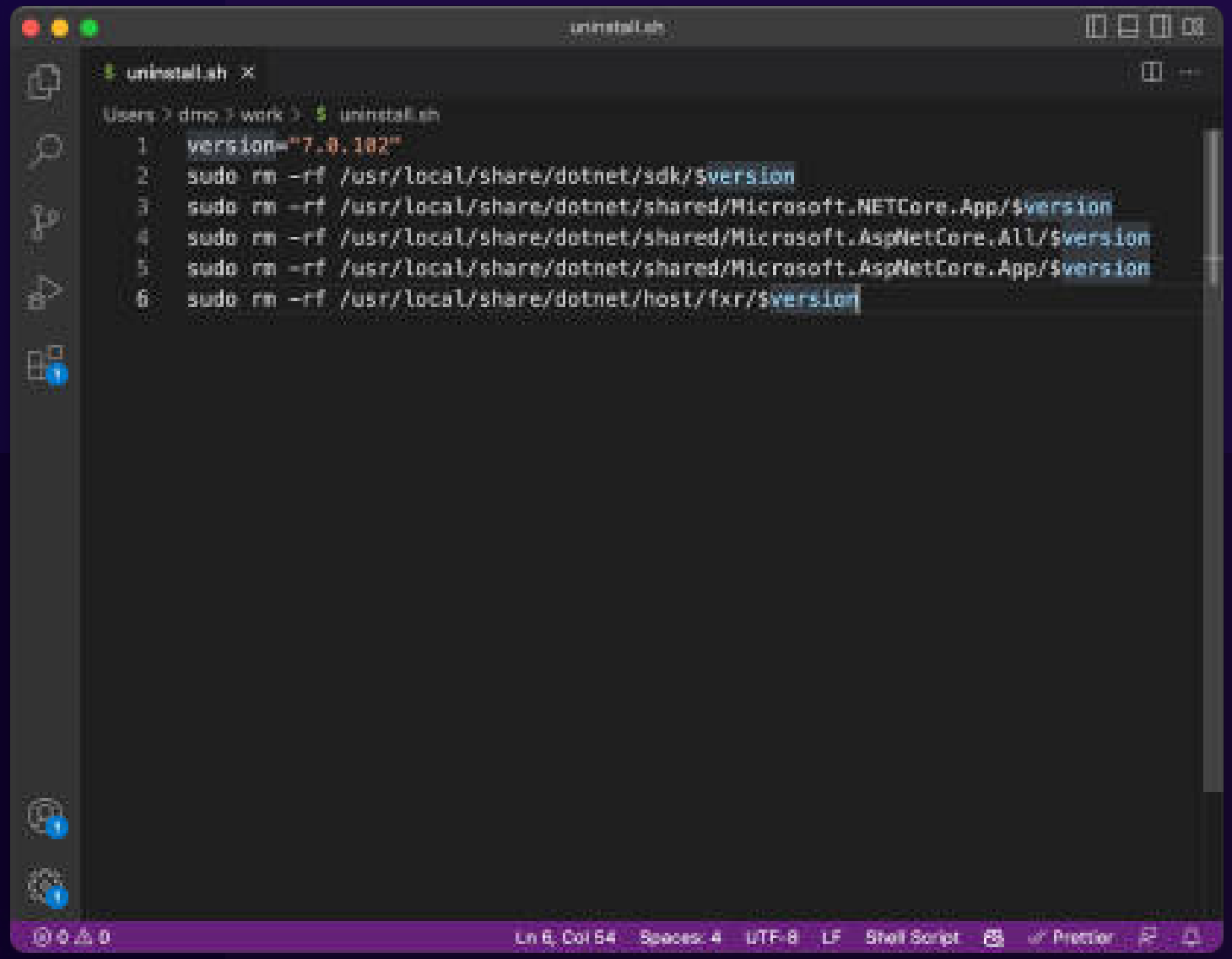
> dotnet nuget list source

> which dotnet



uninstall.sh

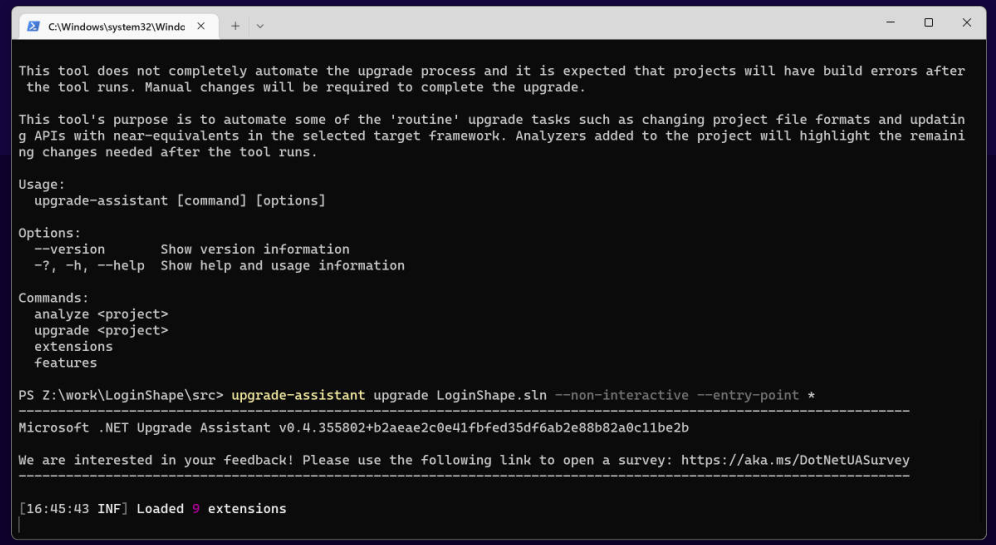
> ./uninstall.sh



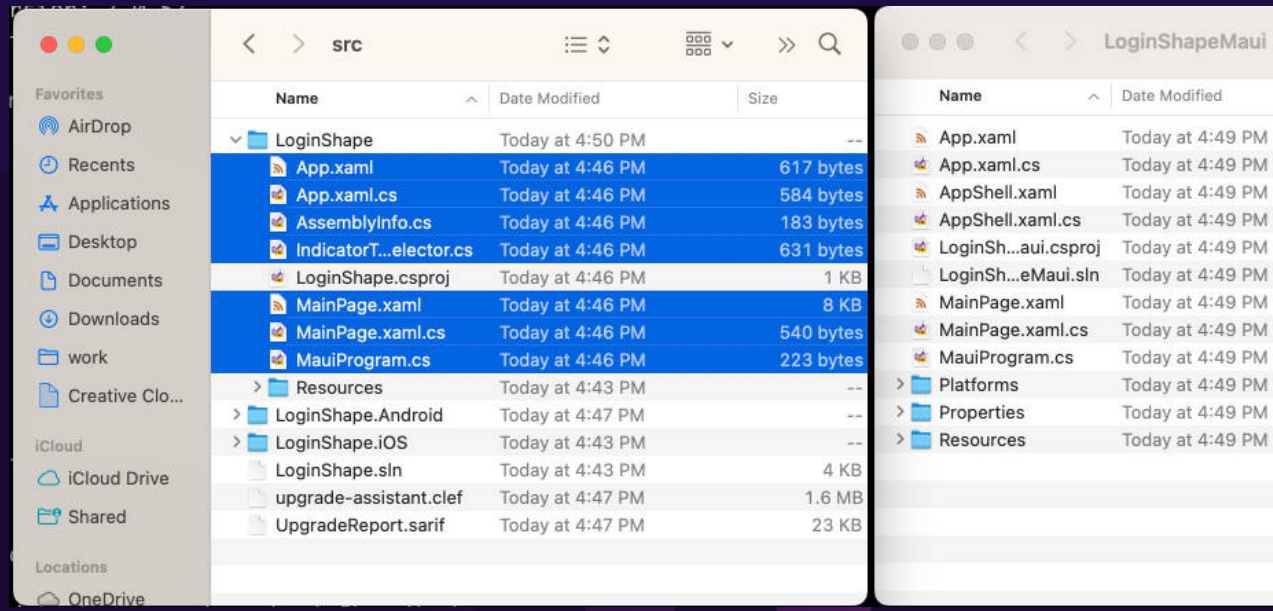
Upgrading from Xamarin

Two Options

- .NET Upgrade Assistant

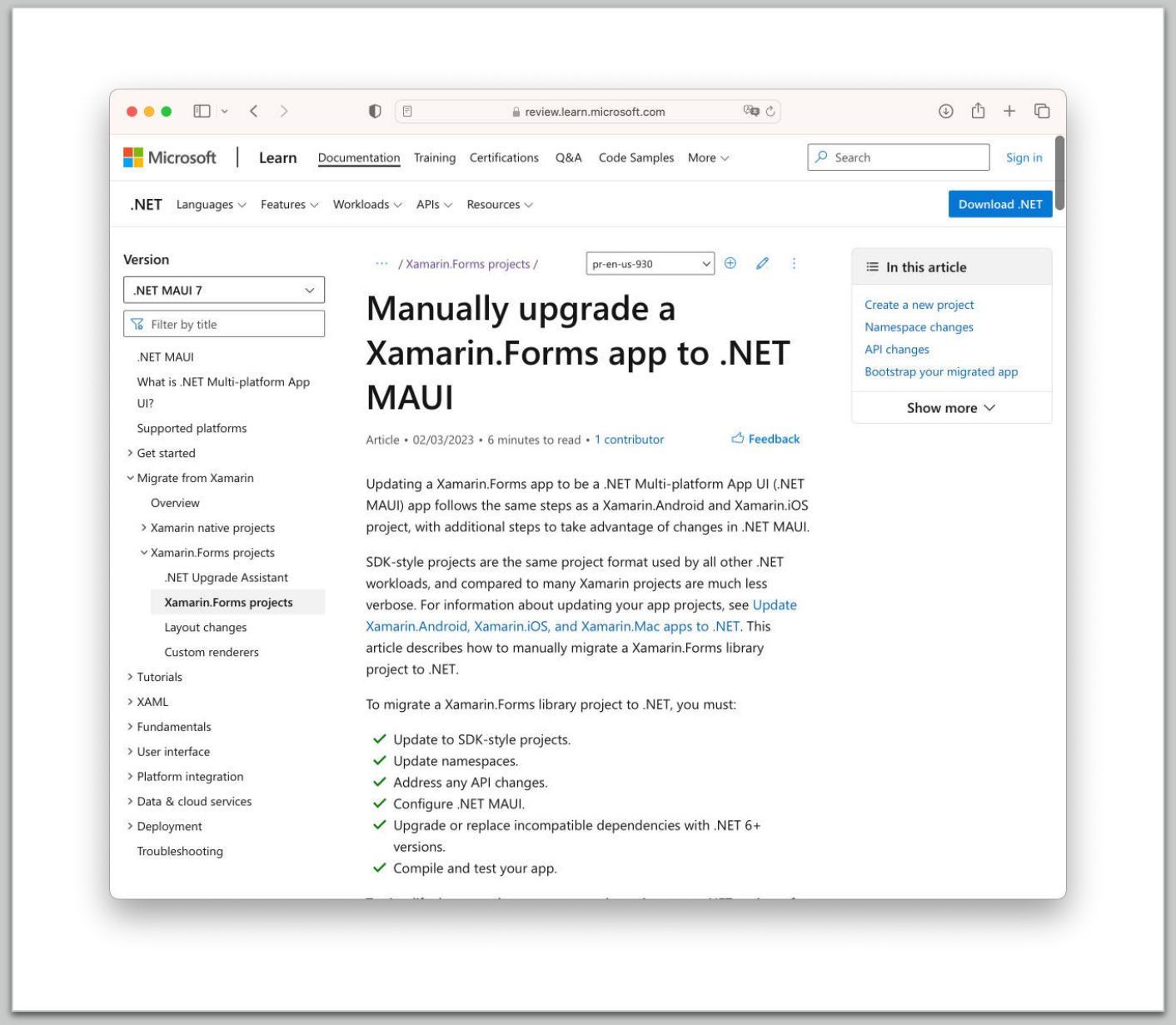


- Copy Pasta

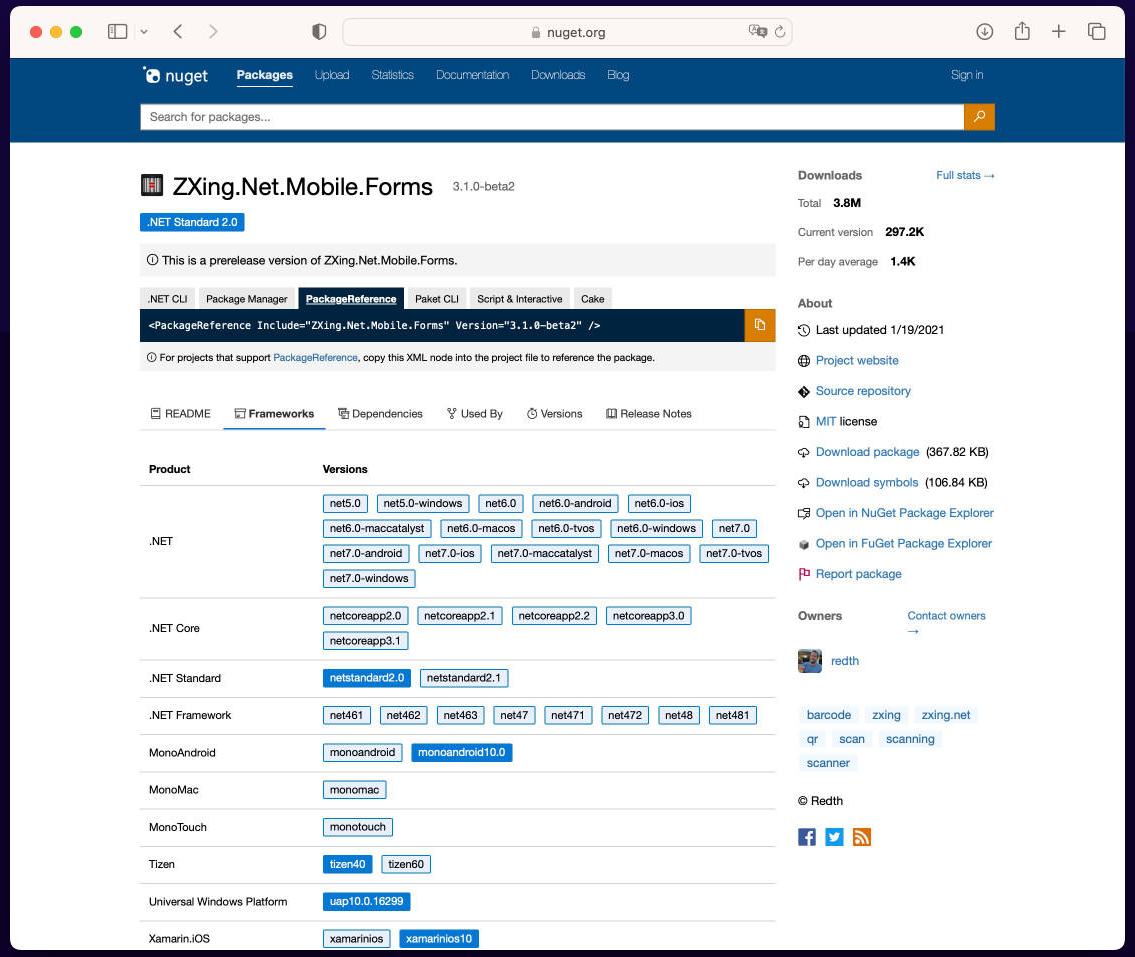


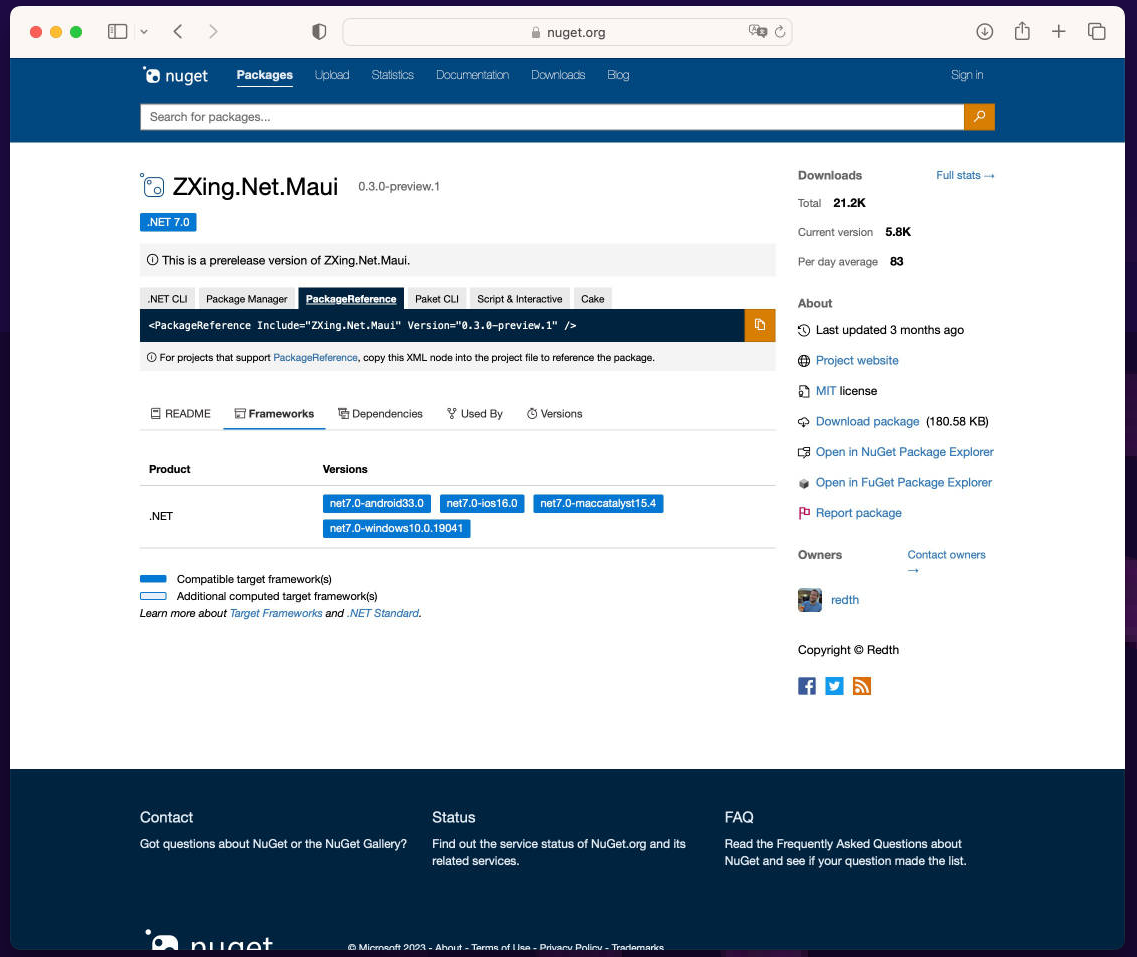
Upgrade tasks

- Update to SDK-style projects.
- Update namespaces.
- Address any API changes.
- Configure .NET MAUI.
- Upgrade or replace incompatible dependencies with .NET 6+ versions.
- Compile and test your app.

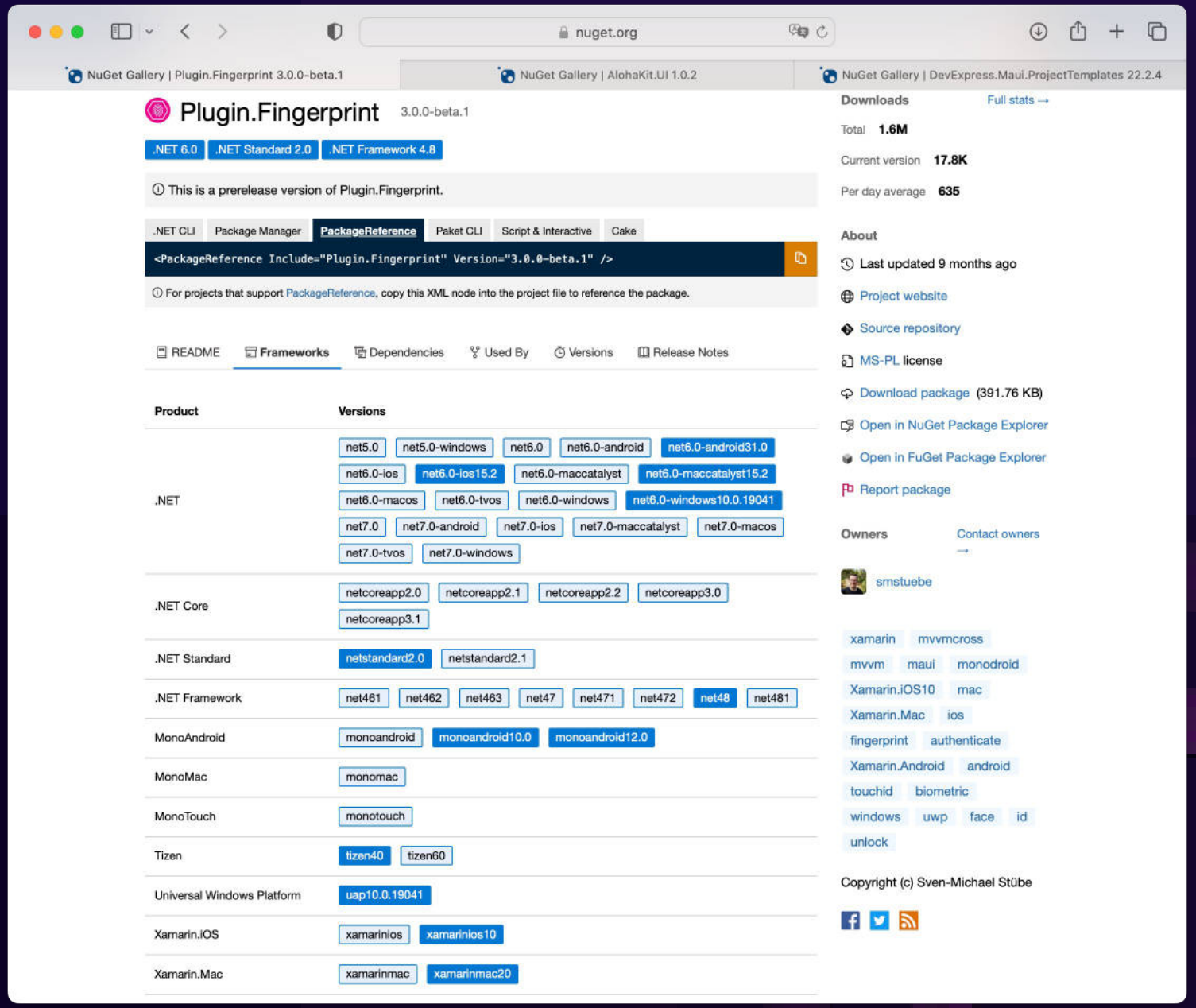


Which Nuget?

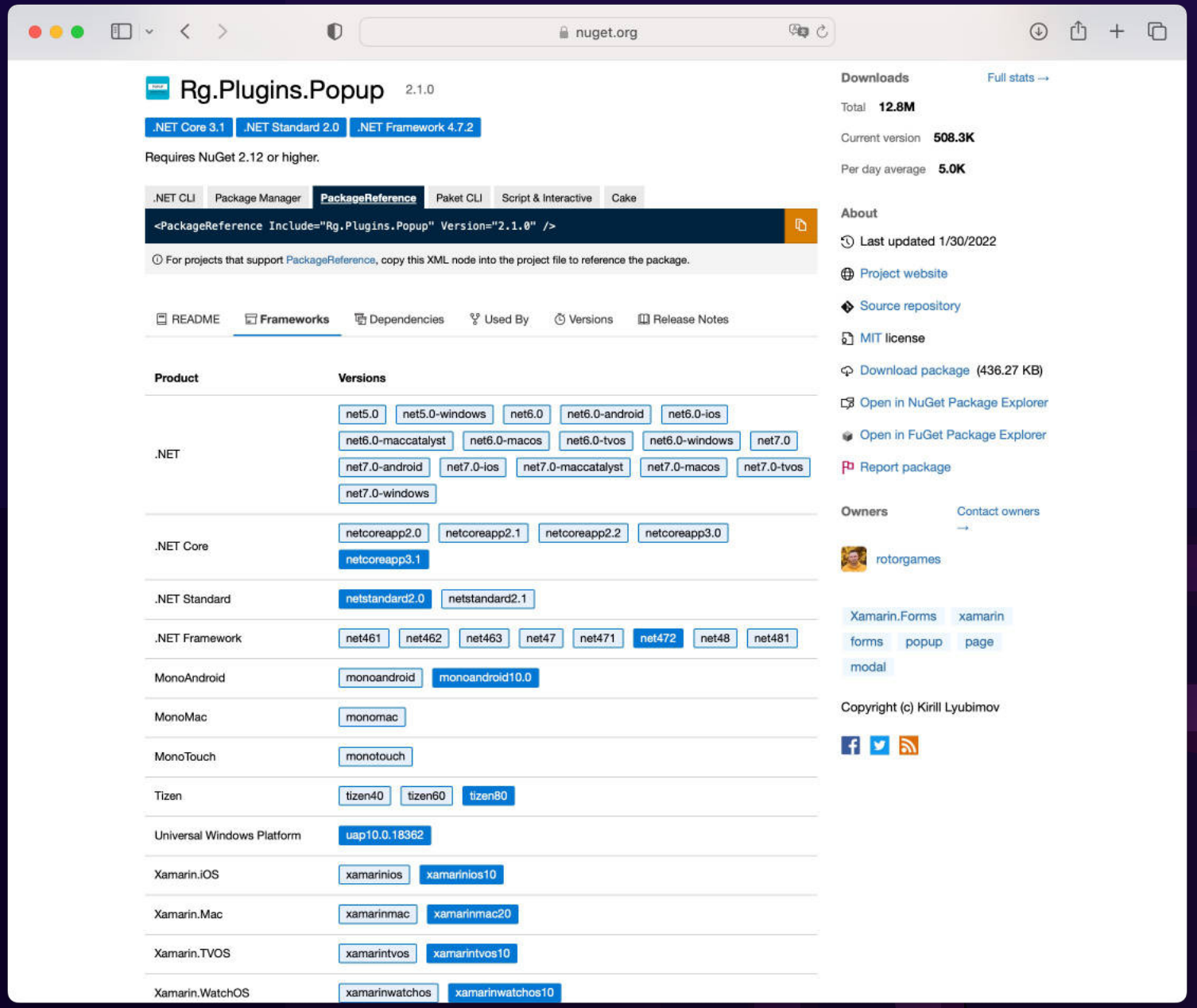

nuget.org
nuget
Packages
Upload
Statistics
Documentation
Downloads
Blog
Sign in
Search for packages...
ZXing.Net.Mobile.Forms
3.1.0-beta2
.NET Standard 2.0
This is a prerelease version of ZXing.Net.Mobile.Forms.
.NET CLI
Package Manager
PackageReference
Paket CLI
Script & Interactive
Cake
<PackageReference Include="ZXing.Net.Mobile.Forms" Version="3.1.0-beta2" />
For projects that support PackageReference, copy this XML node into the project file to reference the package.
README
Frameworks
Dependencies
Used By
Versions
Release Notes
Product
Versions
.NET
net5.0 net5.0-windows net6.0 net6.0-android net6.0-ios net6.0-maccatalyst net6.0-macos net6.0-tvos net6.0-windows net7.0 net7.0-android net7.0-ios net7.0-maccatalyst net7.0-macos net7.0-tvos net7.0-windows
.NET Core
netcoreapp2.0 netcoreapp2.1 netcoreapp2.2 netcoreapp3.0 netcoreapp3.1
.NET Standard
netstandard2.0 netstandard2.1
.NET Framework
net461 net462 net463 net47 net471 net472 net48 net481
MonoAndroid
monoandroid monoandroid10.0
MonoMac
monomac
MonoTouch
monotouch
Tizen
tizen40 tizen60
Universal Windows Platform
uap10.0.16299
Xamarin.iOS
xamarinios xamarinios10
Downloads
Full stats →
Total 3.8M
Current version 297.2K
Per day average 1.4K
About
Last updated 1/19/2021
Project website
Source repository
MIT license
Download package (367.82 KB)
Download symbols (106.84 KB)
Open in NuGet Package Explorer
Open in FuGet Package Explorer
Report package
Owners
Contact owners →
redth
barcode zxing zxing.net qr scan scanning scanner
© Redth


nuget.org
nuget
Packages
Upload
Statistics
Documentation
Downloads
Blog
Sign in
Search for packages...
ZXing.Net.Maui
0.3.0-preview.1
.NET 7.0
This is a prerelease version of ZXing.Net.Maui.
.NET CLI
Package Manager
PackageReference
Paket CLI
Script & Interactive
Cake
<PackageReference Include="ZXing.Net.Maui" Version="0.3.0-preview.1" />
For projects that support PackageReference, copy this XML node into the project file to reference the package.
README
Frameworks
Dependencies
Used By
Versions
Product
Versions
.NET
net7.0-android33.0 net7.0-ios16.0 net7.0-maccatalyst15.4 net7.0-windows10.0.19041
Compatible target framework(s)
Additional computed target framework(s)
Learn more about Target Frameworks and .NET Standard.
Downloads
Full stats →
Total 21.2K
Current version 5.8K
Per day average 83
About
Last updated 3 months ago
Project website
MIT license
Download package (180.58 KB)
Open in NuGet Package Explorer
Open in FuGet Package Explorer
Report package
Owners
Contact owners →
redth
Copyright © Redth
Contact
Got questions about NuGet or the NuGet Gallery?
Status
Find out the service status of NuGet.org and its related services.
FAQ
Read the Frequently Asked Questions about NuGet and see if your question made the list.
© Microsoft 2023 - About - Terms of Use - Privacy Policy - Trademarks

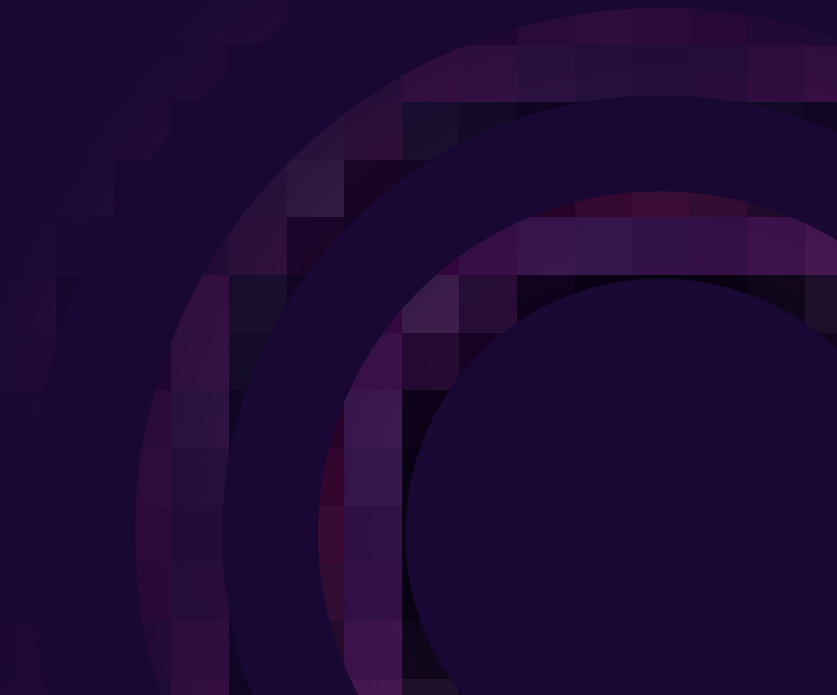
Which Nuget?


nuget.org
NuGet Gallery | Plugin.Fingerprint 3.0.0-beta.1
NuGet Gallery | AlohaKit.UI 1.0.2
NuGet Gallery | DevExpress.Maui.ProjectTemplates 22.2.4
Plugin.Fingerprint 3.0.0-beta.1
.NET 6.0
.NET Standard 2.0
.NET Framework 4.8
This is a prerelease version of Plugin.Fingerprint.
.NET CLI
Package Manager
PackageReference
Paket CLI
Script & Interactive
Cake
<PackageReference Include="Plugin.Fingerprint" Version="3.0.0-beta.1" />
For projects that support PackageReference, copy this XML node into the project file to reference the package.
README
Frameworks
Dependencies
Used By
Versions
Release Notes
Product
Versions
.NET
net5.0 net5.0-windows net6.0 net6.0-android net6.0-android31.0 net6.0-ios net6.0-ios15.2 net6.0-maccatalyst net6.0-maccatalyst15.2 net6.0-macos net6.0-tvos net6.0-windows net6.0-windows10.0.19041 net7.0 net7.0-android net7.0-ios net7.0-maccatalyst net7.0-macos net7.0-tvos net7.0-windows
.NET Core
netcoreapp2.0 netcoreapp2.1 netcoreapp2.2 netcoreapp3.0 netcoreapp3.1
.NET Standard
netstandard2.0 netstandard2.1
.NET Framework
net461 net462 net463 net47 net471 net472 net48 net481
MonoAndroid
monoandroid monoandroid10.0 monoandroid12.0
MonoMac
monomac
MonoTouch
monotouch
Tizen
tizen40 tizen60
Universal Windows Platform
uap10.0.19041
Xamarin.iOS
xamarinios xamarinios10
Xamarin.Mac
xamarinmac xamarinmac20
Downloads
Full stats →
Total 1.6M
Current version 17.8K
Per day average 635
About
Last updated 9 months ago
Project website
Source repository
MS-PL license
Download package (391.76 KB)
Open in NuGet Package Explorer
Open in FuGet Package Explorer
Report package
Owners
Contact owners →
smstuebe
xamarin mvvmcross mvvm maui monodroid Xamarin.iOS10 mac Xamarin.Mac ios fingerprint authenticate Xamarin.Android android touchid biometric windows uwp face id unlock
Copyright (c) Sven-Michael Stübe

Which Nuget?


nuget.org
Rg.Plugins.Popup 2.1.0
.NET Core 3.1
.NET Standard 2.0
.NET Framework 4.7.2
Requires NuGet 2.12 or higher.
.NET CLI
Package Manager
PackageReference
Paket CLI
Script & Interactive
Cake
<PackageReference Include="Rg.Plugins.Popup" Version="2.1.0" />
For projects that support PackageReference, copy this XML node into the project file to reference the package.
README
Frameworks
Dependencies
Used By
Versions
Release Notes
Product
Versions
.NET
net5.0 net5.0-windows net6.0 net6.0-android net6.0-ios net6.0-maccatalyst net6.0-macos net6.0-tvos net6.0-windows net7.0 net7.0-android net7.0-ios net7.0-maccatalyst net7.0-macos net7.0-tvos net7.0-windows
.NET Core
netcoreapp2.0 netcoreapp2.1 netcoreapp2.2 netcoreapp3.0 netcoreapp3.1
.NET Standard
netstandard2.0 netstandard2.1
.NET Framework
net461 net462 net463 net47 net471 net472 net48 net481
MonoAndroid
monoandroid monoandroid10.0
MonoMac
monomac
MonoTouch
monotouch
Tizen
tizen40 tizen60 tizen80
Universal Windows Platform
uap10.0.18362
Xamarin.iOS
xamarinios xamarinios10
Xamarin.Mac
xamarinmac xamarinmac20
Xamarin.TVOS
xamarintvos xamarintvos10
Xamarin.WatchOS
xamarinwatchos xamarinwatchos10
Downloads
Full stats →
Total 12.8M
Current version 508.3K
Per day average 5.0K
About
Last updated 1/30/2022
Project website
Source repository
MIT license
Download package (436.27 KB)
Open in NuGet Package Explorer
Open in FuGet Package Explorer
Report package
Owners
Contact owners →
rotorgames
Xamarin.Forms xamarin forms popup page modal
Copyright (c) Kirill Lyubimov

Layouts



Behavior changes

Layout	Xamarin.Forms	.NET MAUI	Recommendation
All	In certain cases sizing requests would not be honored.	Sizing requests are honored.	
Grid	Columns and rows can be inferred from XAML.	Columns and rows must be explicitly declared.	Add ColumnDefinitions and RowDefinitions.
HorizontalStackLayout		"*AndExpand" has no effect.	
RelativeLayout		Requires the compatibility namespace.	Use Grid.
ScrollView	Arbitrary minimum size, inconsistent sizing	Expands to fill all space	Set a size if within a stack to enable scrolling.
StackLayout	Children can fill space in the stacking direction.	Children are stacked and will go beyond available space.	If you need child views to fill space, change to a Grid.
VerticalStackLayout		"*AndExpand" has no effect.	

Value changes

Property	Xamarin.Forms value	.NET MAUI value
Grid.ColumnSpacing	6	0
Grid.RowSpacing	6	0
StackLayout.Spacing	6	0

StackLayout

- Very simple – stacks in a single direction until all children stacked.
- Does NOT subdivide space
- Remove any "AndExpand" properties which are orthogonal to the stacking direction.

XAML Copy

```
<StackLayout>
    <Label Text="Hello world!"/>
    <Image VerticalOptions="FillAndExpand" Source="dotnetbot.png"/>
</StackLayout>
```

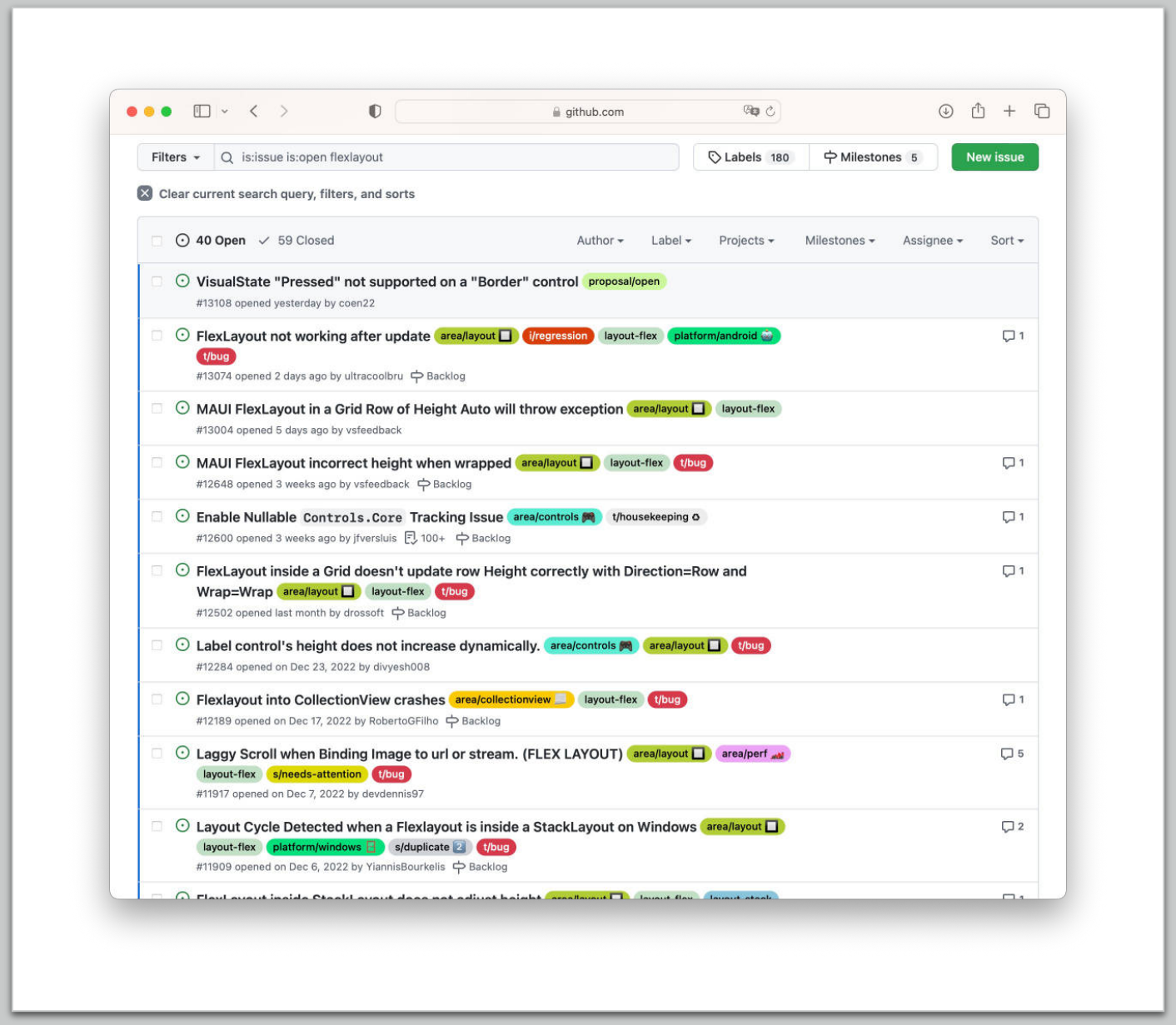
This can be converted to a `Grid` in .NET MAUI:

XAML Copy

```
<Grid RowDefinitions="Auto, *">
    <Label Text="Hello world!"/>
    <Image Grid.Row="1" Source="dotnetbot.png"/>
</StackLayout>
```

FlexLayout

- Be careful



Custom layouts

- Create an ILayoutManager and override:
  - Measure
  - ArrangeChildren

https://github.com/hartez/CustomLayoutExamples

So given this XAML:

```
<local:HorizontalWrapLayout Margin="10" Spacing="5">
    <Button Text="One"></Button>
    <Button Text="Two"></Button>
    <Button Text="Three"></Button>
    <Button Text="Four"></Button>
    <Button Text="Five"></Button>
    <Button Text="Six"></Button>
    <Button Text="Seven"></Button>
    <Button Text="Eight"></Button>
    <Button Text="Nine"></Button>
    <Button Text="Ten"></Button>
</local:HorizontalWrapLayout>
```

You'll see something like this:

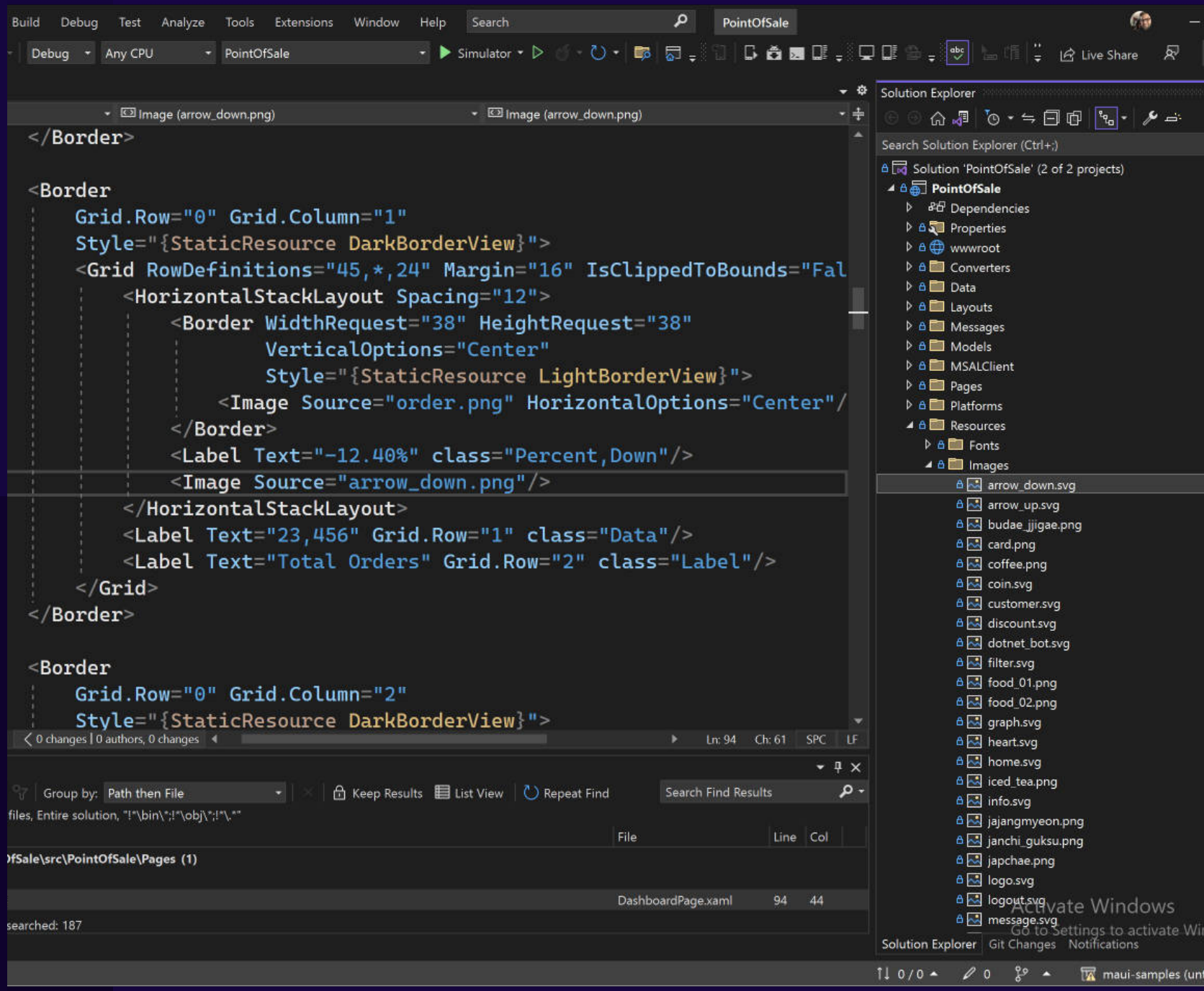


Single Project

Images

- SVGs generate PNGs
- Using the SVG extension still renders a PNG
- Edit the csproj to set:
  - BaseSize
  - Resize
  - Color
  - TintColor

Filenames must be lowercase, start and end with a letter character, and contain only alphanumeric characters or underscores. Thx Google.



App icons and Splash screens

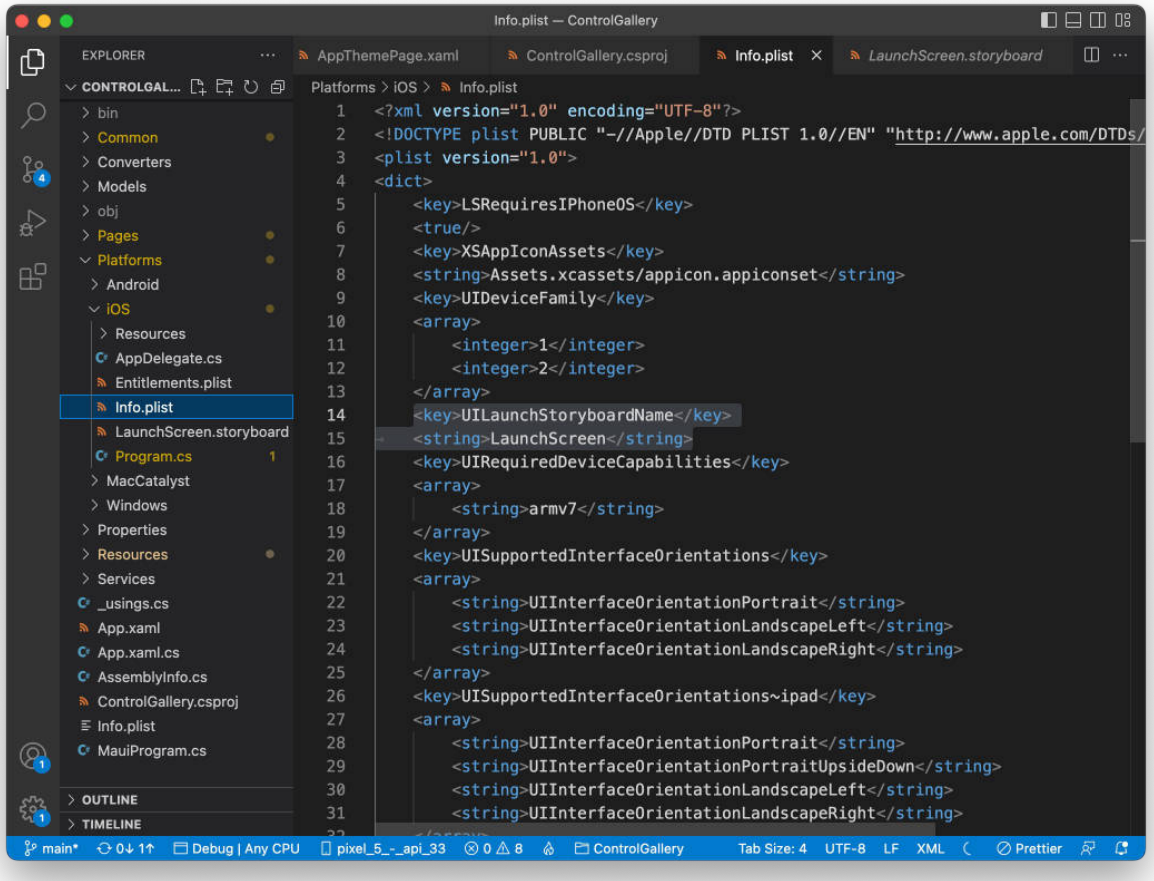
- .NET MAUI makes this easy with :
  - MauiIcon
  - MauiSplashScreen

```
<ItemGroup>
    <!-- App Icon -->
    <MauiIcon
        Include="Resources\AppIcon\appicon.svg"
        ForegroundFile="Resources\AppIcon\appiconfg.svg"
        Color="#512BD4" />

    <!-- Splash Screen -->
    <MauiSplashScreen
        Include="Resources\Splash\splash.svg"
        Color="#512BD4"
        BaseSize="128,128" />
```

App icons and Splash screens

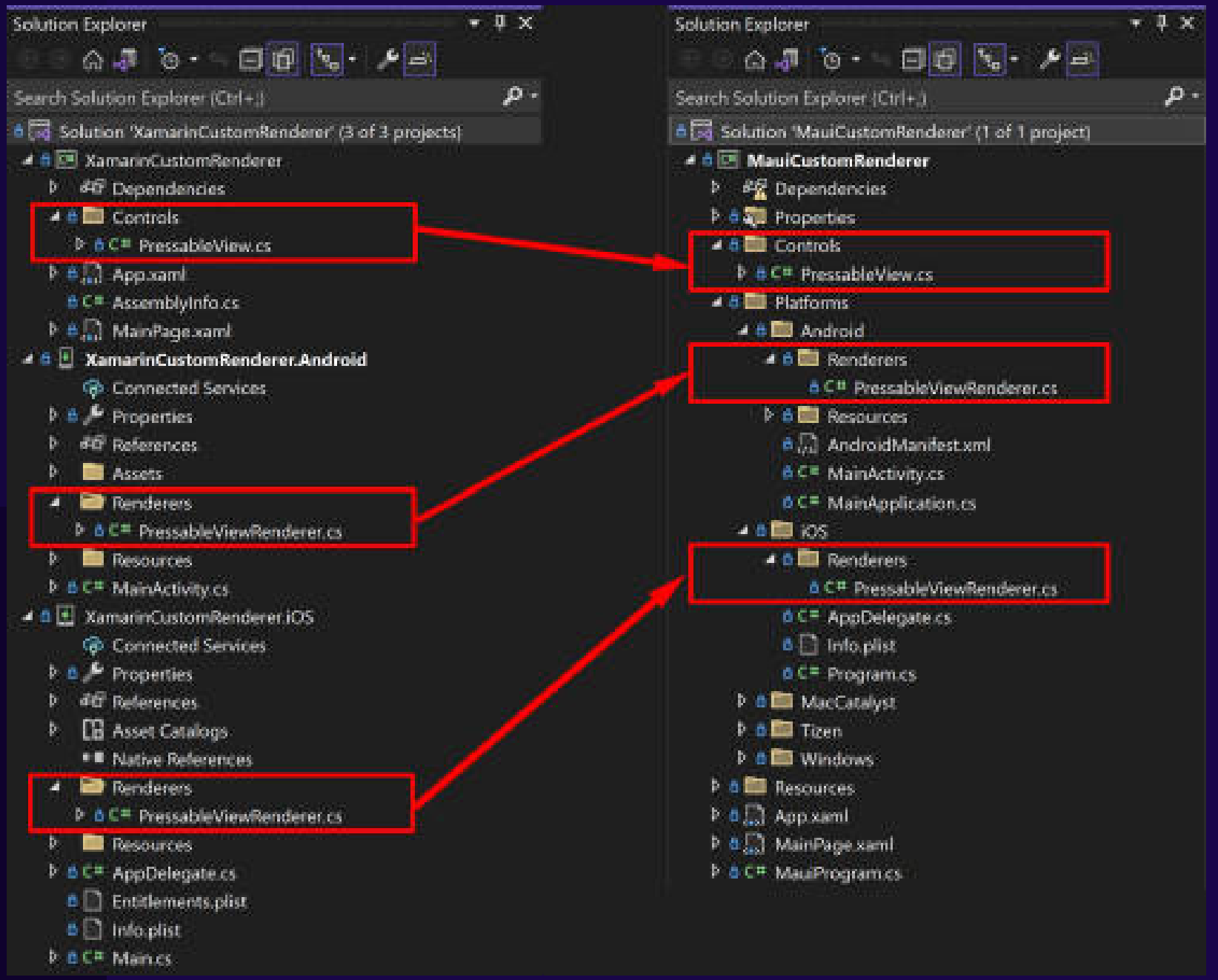
- But you can still make your own “by hand”



Control Customization

Custom Renderers

- Platform folders will multi-target appropriately
- This is optional



Custom Renderers

- Remove [ExportRenderer]
- Configure as a handler
- Use it

C# Copy

```
using Microsoft.Maui.Controls.Compatibility.Hosting;
...

public static class MauiProgram
{
    public static MauiApp CreateMauiApp()
    {
        var builder = MauiApp.CreateBuilder();
        builder
            .UseMauiApp<App>()
            .UseMauiCompatibility()
            .ConfigureFonts(fonts =>
            {
                fonts.AddFont("OpenSans-Regular.ttf", "OpenSansRegular");
                fonts.AddFont("OpenSans-Semibold.ttf", "OpenSansSemibold");
            })
            .ConfigureMauiHandlers((handlers) =>
            {
#if ANDROID
                handlers.AddHandler(typeof(PressableView), typeof(XamarinCustomRend
#elif IOS
                handlers.AddHandler(typeof(PressableView), typeof(XamarinCustomRend
#endif
            });

        return builder.Build();
    }
}
```

Hooks

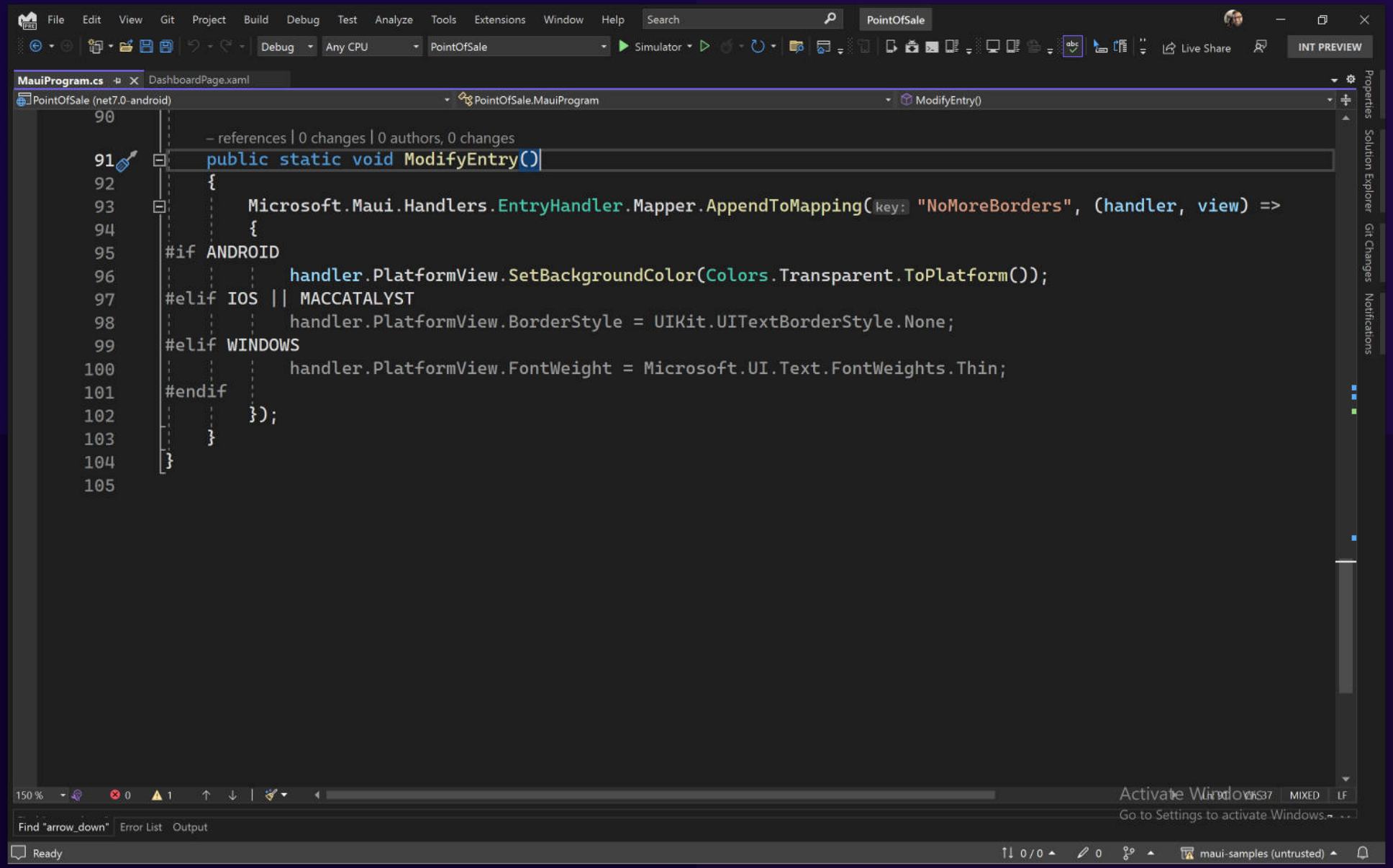
Handlers can be customized to augment the appearance and behavior of a cross-platform control beyond the customization that's possible through the control's API. This customization, which modifies the native views for the cross-platform control, is achieved by modifying the mapper for a handler with one of the following methods:

PrependToMapping, which modifies the mapper for a handler before the .NET MAUI control mappings have been applied.

ModifyMapping, which modifies an existing mapping.

AppendToMapping, which modifies the mapper for a handler after the .NET MAUI control mappings have been applied.

```
1 reference
private static void OverrideHandlers()
{
#if IOS || MACCATALYST
    ButtonHandler.Mapper.ModifyMapping(nameof(Button.ContentLayout), (handler, btn, element) =>
    {
        UpdateContentLayout(handler.PlatformView, (Button)btn);
    });
#endif
}
```


File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search
PointOfSale
Debug
Any CPU
PointOfSale
Simulator
Live Share
INT PREVIEW
MauiProgram.cs
DashboardPage.xaml
PointOfSale (net7.0-android)
PointOfSale.MauiProgram
ModifyEntry()
– references | 0 changes | 0 authors, 0 changes
public static void ModifyEntry()
{
Microsoft.Maui.Handlers.EntryHandler.Mapper.AppendToMapping(key: "NoMoreBorders", (handler, view) =>
{
#if ANDROID
handler.PlatformView.SetBackgroundColor(Colors.Transparent.ToPlatform());
#elif IOS || MACCATALYST
handler.PlatformView.BorderStyle = UIKit.UITextBorderStyle.None;
#elif WINDOWS
handler.PlatformView.FontWeight = Microsoft.UI.Text.FontWeights.Thin;
#endif
});
}
}
150 %
0
1
MIXED
LF
Activate Windows
Go to Settings to activate Windows.
Find "arrow_down"
Error List
Output
Ready
0 / 0
0
maui-samples (untrusted)
Properties
Solution Explorer
Git Changes
Notifications

Hot Reload

.NET Hot Reload

- Implement MetaDataUpdateHandler to trigger UI refresh
- XAML Hot Reload needs to still be enabled

```
#if DEBUG
[assembly: System.Reflection.Metadata.MetadataUpdateHandler(typeof(VolumeMixer.HotReloadService))]
namespace VolumeMixer
{
    3 references | 0 changes | 0 authors, 0 changes
    public static class HotReloadService
    {
#pragma warning disable CS8632 // The annotation for nullable reference types should only be used in
        public static event Action<Type[]?>? UpdateApplicationEvent;
#pragma warning restore CS8632 // The annotation for nullable reference types should only be used in

        0 references | 0 changes | 0 authors, 0 changes
        internal static void ClearCache(Type[]? types) { }
        0 references | 0 changes | 0 authors, 0 changes
        internal static void UpdateApplication(Type[]? types)
        {
            //if ((types != null && types.Length > 0 && types[0] != null))
            //{
            UpdateApplicationEvent?.Invoke(types);
            //}
        }
    }
}
#endif
```

.NET Hot Reload

- Handle event and trigger rebuild of UI

```
2 references | David Ortinau, 2 days ago | 1 author, 1 change
void Build() ...

0 references | David Ortinau, 2 days ago | 1 author, 1 change
protected override void OnNavigatedTo(NavigatedToEventArgs args)
{
    base.OnNavigatedTo(args);

    BindingContext = new MainViewModel();

    Build();

#if DEBUG
    HotReloadService.UpdateApplicationEvent += ReloadUI;
#endif
}
0 references | David Ortinau, 2 days ago | 1 author, 1 change
protected override void OnNavigatedFrom(NavigatedFromEventArgs args)
{
    base.OnNavigatedFrom(args);

#if DEBUG
    HotReloadService.UpdateApplicationEvent -= ReloadUI;
#endif
}

2 references | David Ortinau, 2 days ago | 1 author, 1 change
private void ReloadUI(Type[] obj)
{
    MainThread.BeginInvokeOnMainThread(() =>
    {
        Build();
    });
}
}
```